Quantifying the impact of precision errors on quantum approximate optimization algorithms

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(Received 14 December 2021; accepted 28 April 2025; published 9 June 2025)

The quantum approximate optimization algorithm (QAOA) is a hybrid quantum-classical algorithm that seeks to achieve approximate solutions to optimization problems by iteratively alternating between intervals of controlled quantum evolution. Here, we examine the effect of analog precision errors on QAOA performance from the perspective of both algorithmic training and performance guarantees. Leveraging cumulant expansions, we recast the faulty QAOA as a control problem in which precision errors are expressed as multiplicative control noise and derive bounds on the performance of QAOA. We show using both analytical techniques and numerical simulations that fixed precision implementations of QAOA circuits are subject to an exponential degradation in performance dependent upon the number of optimal QAOA layers and magnitude of the precision error. Despite this significant reduction, we show that it is possible to mitigate precision errors in QAOA via digitization of the variational parameters at the cost of increasing circuit depth.

DOI: 10.1103/PhysRevResearch.7.023240

I. INTRODUCTION

The quantum approximate optimization algorithm (QAOA) [1] provides a hybrid classical-quantum approach to solving combinatorial optimization problems. QAOA has attracted a great deal of attention due to the advent of near- and intermediate-term quantum computing (QC). Furthermore, because of its simplicity, QAOA has been well-suited to both analytical study [1–10] and experimental implementation on a variety of platforms [11–14].

While the idealized QAOA provides performance guarantees [1,5,10], questions remain regarding its practical implementation, namely, its robustness to noise. In recent years, studies have examined QAOA subject to local, environmentally induced Markovian noise [15–18], as well as extensions to spatiotemporally correlated noise mediated by classical environmental fluctuators [19]. Assessments of the impact of systematic error sources, i.e., those that are *not* mediated by an environment (and therefore may persist even in closed systems), have predominately focused on imperfect readout [20]. Investigations of noise due to faulty controls have thus far centered on suppression strategies for static and quasistatic noise [21].

Errors arising from imperfect control can manifest in a variety of ways. In practical implementations of analog QC, such as quantum annealing (QA), on-chip control components (e.g., qubit couplers) used to specify programmable Hamiltonian parameters are subject to variable precision limitations that can drastically differ from the precision of the user-defined problem [22–24]. These limitations lead to *precision errors* that result in a misspecification of the computational problem at the hardware layer that becomes increasingly worse as problem size increases [24]. The impact of precision errors on QA has been highlighted, where, for example, Hamiltonian coupling strengths $J_{ij} \to J_{ij} + \delta J_{ij}$ [25]. The error δJ_{ij} is found to be nonlinear and dependent upon the desired coupling strength. The presence of unmitigated precision errors has proven to be severely harmful [26], leading to an exponential decay in the success probability with problem size and error magnitude [27]. Experimental studies noting the impact of this error source have further sparked the development of mitigation protocols [25,28,29].

In contrast to existing QA systems, currently available digital QC systems are not limited by precision errors. This is in part due to their restriction to low-depth quantum algorithms consisting of tens of qubits and their predominate use of off-chip control hardware. Next generation intermediate scale quantum systems, however, are quickly trending toward systems consisting of thousands of qubits capable of implementing larger optimization problems that require high-depth circuits. These systems will demand scalable, fast, flexible,

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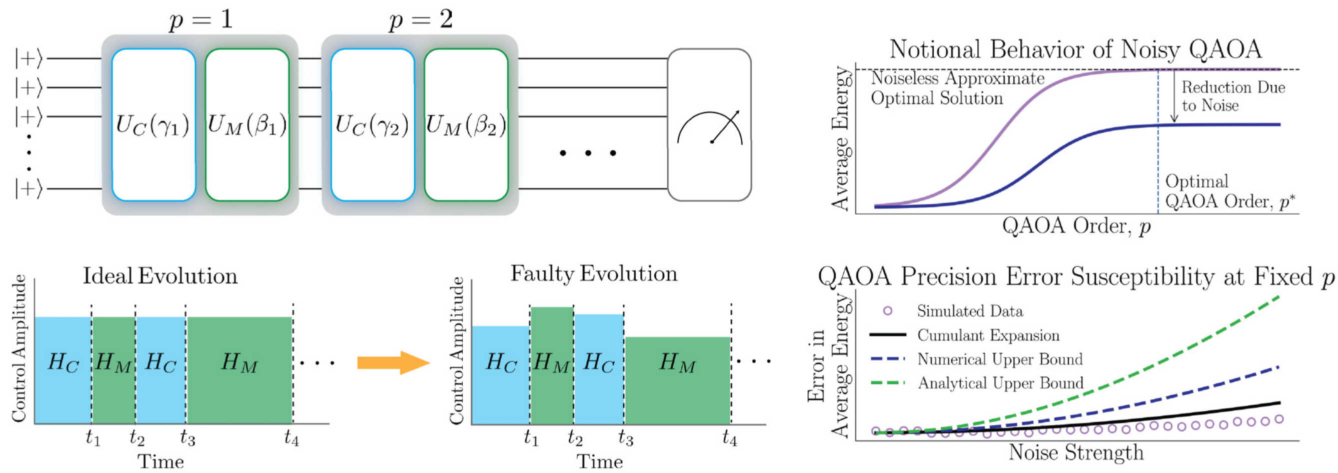

FIG. 1. Schematic figure describing the effect of faulty QAOA evolution resulting from precision errors. Top left: Schematic of typical QAOA *Ansätze*. Bottom left: Perspective of precision errors on QAOA evolution. Ideally, each Hamiltonian is implemented with a specific amplitude and duration. Precision errors in the variational parameters can be modeled as amplitude errors in the QAOA control evolution. Top right: Notional behavior of QAOA in the ideal and noisy setting. As the QAOA order is increased, the algorithm performance improves. Ultimately, performance plateaus at the approximate optimal solution, where an optimal QAOA order can be determined. The presence of noise alters this behavior, leading to a reduction in the optimal approximate solution. Bottom right: Utility of cumulant-based perspective on faulty QAOA illustrated for notional QAOA performance in the presence of precision errors. The cumulant expansion provides an analytical expression for the approximate dynamics of the faulty QAOA, as well as error bounds on QAOA performance.

and high-precision control composed of on-chip components to reduce power, size, and cost requirements while enabling high-fidelity gates [30–32]. If advances in classical control systems do not remain commensurate with progress in quantum processors, digital QC could be subject to similar limitations to those found in QA. Namely, precision errors could rapidly emerge as a dominant error source for both physical and virtual gate operations in control hardware relevant to superconducting qubit, trapped ion, and semiconductor qubit systems [31].

From an algorithmic perspective, precision errors in digital QC systems can manifest as misspecification of cost Hamiltonian parameters in QAOA. Alternatively, constraints in control hardware precision can propagate to the variational parameters $\{\theta_i\}$ and appear as imperfections in the QAOA evolution. An example is $\theta_i \to \theta_i + \delta\theta_i$,, where $\delta\theta_i$ is dependent upon the precision limits of the qubit control chain. Since QAOA evolution is generated by a product of the variational parameters and the Hamiltonian, the resulting errors are equivalent despite their distinct origin.

Despite the intimate relationship between QAOA and QA [1,33–35], the overall impact of precision errors on QAOA performance is not well understood. It is unclear if the behavior observed for QA is indeed a bad omen for large-scale QAOA. Furthermore, if an equivalent susceptibility exists, how does one address this error in near-term systems, where fault-tolerant quantum error correcting codes cannot be exploited to combat digitization errors [36]? In particular, can software approaches be designed to off-load burdens on control hardware and enable robustness?

In this study, we perform a comprehensive analysis of precision errors in QAOA and propose a software-based approach for combating these errors in existing and next-generation quantum computing systems. We consider the regime where precision errors are the dominant error source and model them as errors in the variational parameters. We show that such errors can equivalently be modeled as faulty controls and investigate the impact of both static and time-varying errors. While static errors can be addressed by variational optimization, the latter requires a more sophisticated approach. To this end, we propose a compilation scheme that relies on the digitization of the QAOA variational parameters.

Our approach leverages concepts from quantum control theory to analytically estimate the contribution of precision errors to the QAOA evolution. This approach paves the way for the development of bounds on approximation ratios and training error. While focused on precision errors, the analytical tools developed in this study are readily extensible to broader classes of spatiotemporally correlated noise models [37–42].

Through our analysis, we argue that any fixed precision implementation of QAOA is destined to detrimentally affect the success of QAOA, which represents a fundamental limitation to the scalability of the algorithm and, more broadly, Ising machines. In particular, we find an *exponential reduction* in success probability with increasing problem size and error magnitude. This observation is displayed through numerical investigations of the QAOA variant of Grover's search and the one-dimensional transverse-field Ising model. Despite the apparent "doomsday scenario" for QAOA, we show that it is possible to mitigate precision errors in QAOA via our digitization strategy. A summary of our study is illustrated by Fig. 1.

II. SUMMARY OF MAIN RESULTS

Variational training parameters in QAOA are susceptible to precision errors that potentially threaten the success of the algorithm. We model precision errors via a control noise contribution to the QAOA Hamiltonians. This results in a misspecification of the QAOA control amplitudes, as depicted in the bottom left panel of Fig. 1. Mathematically, parameter misspecifications are captured by time-dependent stochastic error functions that are characterized by their mean and two-point autocorrelation function (see Sec. III B for further details). Upon subjecting QAOA to stochastic precision errors, we observed an exponential decrease in the achievable approximate optimal solution for QAOA with increasing noise strength and optimal QAOA order [see Fig. 1 (bottom right) for an illustration of this effect].

Using concepts from quantum control theory, we quantify the effect of precision errors on expectation values and unitary operators. Recent work in quantum control has sought to utilize cumulant expansions to assess interplay between control schemes and temporally correlated noise processes [37–39]. Inspired by this work, we leverage a cumulant expansion to examine the dynamics of a system subject to a control designated by the noiseless QAOA protocol and noise produced by a control error Hamiltonian representing precision errors. The cumulant-based approach affords a novel perspective on faulty QAOA dynamics that facilitates the development of bounds on various QAOA-relevant quantities. In particular, our approach allows for the development of bounds on approximation ratios and parameter training error. Figure 1 (bottom right) illustrates the utility of our approach for assessing QAOA susceptibility to precision errors. The cumulant expansion yields expressions for the approximate dynamics of the noisy QAOA. Upper bounds on QAOA-relevant quantities (e.g., the error in average energy) are determined via the cumulant approach. Together, these expressions capture the relative behavior of QAOA as a function of noise strength and QAOA order.

To benchmark our analysis we numerically study the performance of QAOA in the presence of precision errors in two specific implementations:

(i) *Grover earch.* We implement the analytically obtained optimal Grover QAOA for identifying a marked state $|0\rangle$ from a database of $N = 2^n$ entries [5]. The optimal circuit has $p^* \sim \sqrt{N}$ layers.

(ii) *Ising instances.* We utilize optimal circuits to prepare n-qubit Greenberger-Horne-Zeilinger (GHZ) states using QAOA with an Ising Hamiltonian on a ring geometry [43]. The optimal circuit has $p^* = n/2$ layers.

In both cases, p^* is the optimal number of QAOA layers and n indicates the number of qubits. In the case of Grover's search, p^* denotes the query complexity, or minimum number of QAOA layers to achieve the first maximum in the objective function. In the case of the Ising instances, it corresponds to the minimum number of QAOA layers to achieve perfect GHZ state preparation.

We numerically simulate the QAOA circuits for these two problem instances and calculate the measured errors in the cost observable as well as the average distance between the noisy and perfect unitary operators. The precision errors are drawn from a normal distribution, $\mathcal{N}(\eta, \sqrt{\Gamma})$ with mean η and variance Γ. Specifically, we consider two representative limiting cases: (a) coherent errors, $\Gamma = 0$, drawn from $\mathcal{N}(\eta, 0)$, and (b) stochastic errors, $\eta = 0$, drawn from $\mathcal{N}(0, \sqrt{\Gamma})$.

Finally, let us summarize the main numerical results as well as the analytical bounds obtained on the faulty QAOA evolution as a result of precision errors. Let us consider a QAOA circuit with a total run-time T (and layers p) and perfect implementation leading to the state $\rho_0(T)$.

Expectation values. The noise-averaged error in the measured expectation value of any observable, O, has the following upper bound:

$$|\overline{\Delta \langle O(T)\rangle}| \leqslant (e^{\|\mathcal{C}(T)\|_\infty} - 1)\|O\rho_0(T)\|_1, \tag{1}$$

where $\mathcal{C}(T)$ [defined in Eq. (C3)] denotes the cumulant expansion for the error operator $\Lambda(T)$ that depends on the noise as well as the observable, O [see Eq. ((21). The operator $\mathcal{C}(T)$ is dependent upon the specifications of the problem; however, it can be shown to scale as $\|\mathcal{C}(T)\|_\infty \lesssim \mathcal{O}(\eta p^*)$ and $\|\mathcal{C}(T)\|_\infty \lesssim \mathcal{O}(\Gamma p^*)$ for constant and stochastic precision errors, respectively. The numerical investigations of QAOA circuits for both examples considered indicate the following scaling for the error in expectation value of the cost function Hamiltonian, H_C, for weak noise:

$$|\Delta H_C| \approx \langle H_C\rangle_0\,(1 - \exp[-\chi_H(p^*)]), \tag{2}$$

$$\text{with } \chi_H(p^*) \equiv \begin{cases} \chi(\Gamma p^*), & \text{stochastic error} \\ \chi(\eta^a p^*), & \text{coherent error,} \end{cases} \tag{3}$$

where $\langle H_C\rangle_0$ is the noiseless expectation value and the exact functional form of $\chi_H(p^*)$ depends on the details of the QAOA algorithm. The effects of the coherent error turn out to be dependent on the algorithm, with $a = 1$ for the Grover search and $a = 2$ for the Ising problem.

Unitary operators. We obtain a bound on the operator norm of the difference between the perfect and faulty unitary operators for the QAOA evolution via the Frobenius norm,

$$\overline{\|U - U_0\|}_\infty^2 \leqslant \overline{(\Delta U)^2} \leqslant 2(e^{\|\mathcal{C}_U(T)\|_\infty} - 1), \tag{4}$$

where $\Delta U = \|U - U_0\|_2$ and $\mathcal{C}_U(T)$ denotes the cumulant expansion for the error operator. [Note that the difference between unitaries has a direct relationship with the error in the expectation value; see Eq. (E5).] Numerically, we evaluate the ∞-norm difference between the QAOA evolution operators for the Grover search algorithm. For weak noise and small number of layers, p, the average difference scales as

$$\overline{\|U - U_0\|}_\infty \propto \begin{cases} \sqrt{\Gamma} p, & \text{[stochastic error]} \\ \eta p, & \text{[coherent error].} \end{cases} \tag{5}$$

For a large number of layers, or equivalently large noise, the difference saturates to a maximum value of 2.

Training error. We derive bounds on the error in training variational parameters for gradient-based and closed-loop optimization routines. The former follows from the absolute error in the gradient of the expectation value, which can be bounded by expressions proportional to the operator norm (largest singular value) of the error operator $\Lambda(T)$ and its gradient. We estimate training error in closed-loop optimization via bounds on the mean-square error (MSE) in the expectation value of H_C. In the case of constant errors, the training error is bounded by

$$\mathrm{MSE}(H_C(T)) \leqslant 2C_{\max}^2\big(1 + 2h_E^2\big), \tag{6}$$

where $C_{\max}$ is the operator norm of H_C (or, equivalently, the maximum value of the cost function) and h_E effectively represents the accumulated precision error [see Eq. (31)]. Similar expressions that depend more generically on the operator norm of $\Lambda(T)$ are obtained for stochastic precision errors.

Approximation ratio. The bound on the noise-averaged error in the expectation value naturally leads to the bound on the error in the optimal approximation ratio

$$|\overline{\Delta\epsilon^*}| \leqslant \frac{1}{C_{\max}}(e^{\|\mathcal{C}_O(T^*)\|_\infty} - 1)\|H_C\,\rho_0(T^*)\|_1. \tag{7}$$

The algorithmic run-time T^* denotes the time required to achieve the noiseless optimal approximation ratio ϵ^*.

Through our analytical and numerical analysis, we find that stochastic precision errors yield exponentially increasing error in expectation values and distance between unitaries. This observation subsequently propagates to parameter training and approximation ratios. Despite the detrimental nature of precision errors, we devise a strategy for effectively addressing them by digitizing the variational parameters in a binary representation and implementing each QAOA evolution operator as a composite operator. This strategy relies on each constituent operator possessing a greater precision than the desired precision of the QAOA. More precisely, for a p-layer QAOA to achieve a desired accuracy of $2^{-\epsilon}$, the precision error can be at most $\sim\epsilon/\sqrt{p}$. Provided that such stipulations can be met, the digitization strategy ultimately requires an increase in the circuit depth by a factor $\propto|\log(\epsilon/\sqrt{p})|$. Digitization is not required for constant coherent errors, where the over- or under-rotation can be accounted for (1) by updating the optimized angles exactly by the coherent error or (2) intrinsically by a closed-loop classical optimization routine used for parameter training.

III. THE QUANTUM APPROXIMATE OPTIMIZATION ALGORITHM WITH PRECISION ERRORS

A. Noiseless QAOA

In combinatorial optimization, approximation algorithms seek solutions with provable guarantees on the distance between the value of the returned solution and the global optimum. More concretely, consider an objective function $C : \{0,1\}^n \to \mathbb{R}$ which is to be optimized. An ϵ^*-approximation algorithm with high probability achieves a solution x^* such that

$$\frac{C(x^*)}{C_{\max}} \geqslant \epsilon^*. \tag{8}$$

Under these conditions, the algorithm insures with high probability that all solutions are within the approximation ratio ϵ^* of the maximum $C_{\max}$.

The QAOA generates approximate solutions by stroboscopically alternating between two intervals of controlled evolution. In QAOA, these evolutions are generated by a mixing Hamiltonian H_M and a problem Hamiltonian H_C. The mixing Hamiltonian can take various forms; however, typically, it assumes the form of a global transverse field

$$H_M = \sum_{j=1}^{n} X_j. \tag{9}$$

Other mixing Hamiltonians that account for problem constraints have also been examined [9].

The problem Hamiltonian is constructed by translating the classical cost function of n-binary variables into a Hamiltonian of n qubits. This is accomplished by introducing the variable transformation $x_i = (1 - z_i)/2$, where $z_i \in \{-1, 1\}$. The cost function $C(z)$ is then transformed into H_C by replacing z_i with the Pauli operator σ_i^z. The resulting problem Hamiltonian is an Ising Hamiltonian defined on the edges of the underlying graph of the cost function $C(z)$. Note that H_C acts diagonally on computational basis states $|z\rangle$, i.e.,

$$H_C\,|z\rangle = C(z)\,|z\rangle \tag{10}$$

The QAOA is formally implemented via a time-dependent Hamiltonian

$$H_0(t) = [1 - s(t)]H_M + s(t)H_C, \tag{11}$$

where

$$s(t) = \begin{cases} 1 & : \quad t \in \Delta t_j^C \\ 0 & : \quad t \in \Delta t_j^M, \end{cases} \tag{12}$$

defines a stroboscopic function that toggles between weighted time evolutions generated by H_M and H_C. The time intervals $\Delta t_j^C = [t_{2j-2}, t_{2j-1}]$ and $\Delta t_j^M = [t_{2j-1}, t_{2j}]$, $j = 1, \ldots, p$, define the periods during which H_C and H_M, respectively, govern the evolution. The dynamics generated by H_0 can be expressed as

$$\begin{aligned} U_0(T,0) &= \mathcal{T}_+ e^{-i\int_0^T H_0(t)dt} \\ &= U_M(\beta_p)U_C(\gamma_p)\cdots U_M(\beta_1)U_C(\gamma_1), \\ &= U_0^{(p)}(\boldsymbol{\gamma},\boldsymbol{\beta}), \end{aligned} \tag{13}$$

where $\mathcal{T}_+$ denotes the time-ordering operator. The variational parameters $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_p)$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)$ control the time intervals over which H_M and H_C are applied. Each (γ_j, β_j) contains real numbers that parametrize the evolution operators $U_M(\beta) = e^{-i\beta H_M}$ and $U_C(\gamma) = e^{-i\gamma H_C}$. Furthermore, they characterize the effective total time (or "angle budget") $T = \sum_{j=1}^p |\gamma_j| + |\beta_j|$ [33–35].

QAOA evolution is applied to a state initialized in an equal superposition state between all computational basis states, or equivalently the ground state of $-H_M$. The initial state is explicitly given by $\rho(0) = |+\rangle\langle+|^{\otimes n}$, where $|+\rangle = 1/\sqrt{2}(|0\rangle + |1\rangle)$ is defined by the single-qubit computational basis states $\{|0\rangle, |1\rangle\}$. The state resulting from the QAOA evolution is then given by

$$\rho_0(T) = U_0^{(p)}(\boldsymbol{\gamma},\boldsymbol{\beta})\,\rho(0)\,U_0^{(p)}(\boldsymbol{\gamma},\boldsymbol{\beta})^\dagger. \tag{14}$$

The variational parameters are optimized using a classical optimization routine. While there are many options for this procedure [44–47], all of them seek to obtain an approximate solution to the optimization problem by maximizing

$$F(\boldsymbol{\gamma}, \boldsymbol{\beta}) = \langle H_C \rangle_{\boldsymbol{\gamma},\boldsymbol{\beta}} = \mathrm{Tr}[\rho_0(T) H_C], \tag{15}$$

the average energy with respect to the problem Hamiltonian. One can define the approximation ratio for the QAOA with respect to Eq. (15) as

$$\epsilon \equiv F(\boldsymbol{\gamma}, \boldsymbol{\beta})/C_{\max}. \tag{16}$$

The QAOA is now said to be an ϵ^*-approximation algorithm if $\epsilon \geqslant \epsilon^*$. The approximation condition is guaranteed if the resulting quantum state $\rho_0(T)$, when measured in the computational basis, is highly concentrated on solutions that are ϵ^*-approximately optimal [1].

B. Precision errors

Control Hamiltonians are designed to implement a particular quantum operation. However, physical realizations of these Hamiltonians are often accompanied by undesired control-dependent additive or multiplicative noise [39,40,48]. As a result, unwanted system dynamics generated by control parameter inaccuracies lead to control errors. Precision errors constitute a particular type of control error that, while being associated with the misspecifications of variational parameters, also can be treated as control amplitude errors in the QAOA evolution. It is this quantum control point of view that we utilize to examine precision errors in the QAOA setting.

Precision errors can be modeled in QAOA by unintended deviations in the variational parameters. This is equivalent to introducing control noise into the QAOA evolution, such that the faulty dynamics are governed by

$$H(t) = H_0(t) + H_E(t). \tag{17}$$

The dynamics are now described by the ideal QAOA and an additional error Hamiltonian. Here, we focus on uniform precision errors via

$$H_E(t) = \eta_M(t)[1 - s(t)]H_M + \eta_C(t)s(t)H_C. \tag{18}$$

In the context of misspecification of the Hamiltonian, this can be thought of as an equivalent error among all terms in either the cost or mixer Hamiltonian. We generalize to nonuniform errors in Appendix A and show that the error dynamics (see Sec. IV) are only slightly modified. Importantly, it does not alter the overall scaling behavior for the noise models considered. As such, the uniform error case is sufficient to investigate the effect of precision errors on QAOA.

The error functions $\eta_M(t)$ and $\eta_C(t)$ capture the time-dependent control errors for the mixer and problem Hamiltonians, respectively. The evolution generated by Eq. (17) is that of a faulty QAOA evolution $U_0^{(p)}(\tilde{\boldsymbol{\gamma}}, \tilde{\boldsymbol{\beta}})$, where $\tilde{\gamma}_j = \gamma_j(1 + \eta_{C,j})$ and $\tilde{\beta}_j = \beta_j(1 + \eta_{M,j})$ are implemented rather than the ideal evolution of Eq. (13). So far our model describes a general class of multiplicative control errors, i.e., errors that are generated by a faulty control Hamiltonian where the error magnitude scales proportionally with the control.

The statistical properties of the error functions further specify the precision error characteristics. Namely, we model the error functions as stationary, Gaussian random variables with mean $\overline{\eta_\mu(t)} = \eta_\mu$ and time-dependent two-point correlation functions

$$\overline{\eta_\mu(t_1)\eta_\nu(t_2)} = \delta_{\mu\nu}\Gamma_\mu(t_1 - t_2), \tag{19}$$

where $\Gamma_\mu(t_1 - t_2)$ describes the autocorrelation function for $\mu, \nu \in \{M, C\}$. Note that $\overline{\cdots}$ denotes classical ensemble averaging. Below, we will focus on two cases: (1) constant coherent errors, where $\Gamma_\mu \equiv \eta_\mu^2$, and (2) stochastic errors, where $\Gamma_\mu(t_1 - t_2) = \Gamma_\mu \delta(t_1 - t_2)$. Together, these scenarios capture two extremes of precision errors, each uniquely detrimental to QAOA with distinct scaling behavior in QAOA performance as a function of p.

IV. ERROR DYNAMICS

A. Rotating-frame dynamics

The dynamics of $H(t)$ are examined by moving into the rotating reference frame with respect to the control—commonly referred to as the *toggling frame*—and employing time-dependent perturbation theory. In particular, we define the rotated error Hamiltonian $\tilde{H}_E(t) = U_0(T,t)H_E(t)U_0^\dagger(T,t)$, which can be expressed as

$$\tilde{H}_E(t) = \eta_M(t)[1 - s(t)]\tilde{H}_M(t) + \eta_C(t)s(t)\tilde{H}_C(t). \tag{20}$$

The mixer and problem Hamiltonian in the rotated frame are given by $\tilde{H}_M(t) = U_0(T,t)H_M U_0^\dagger(T,t)$ and $\tilde{H}_C(t) = U_0(T,t)H_C U_0^\dagger(T,t)$, respectively. Note that the rotated error Hamiltonian is expressed in terms of a "reverse" interaction picture with respect to the ideal QAOA evolution. As a result, the total dynamics evolution operator $U(T) = \mathcal{T}_+ e^{-i\int_0^T H(t)dt}$ is factorized into the product $U(T) = \tilde{U}_E(T)U_0(T)$, where $\tilde{U}_E(T) = \mathcal{T}_+ e^{-i\int_0^T \tilde{H}_E(t)dt}$ and $U_0(t)$ is the ideal evolution generated by Eq. (11).

B. Time-dependent dynamics of Observables

Using the toggling-frame representation, we investigate the effect of precision errors on the time-dependent expectation value of an observable O. Expectation values, such as the average energy with respect to H_C, are commonly utilized in QAOA to train variational parameters. Furthermore, they are employed to evaluate algorithmic performance via the approximate ratio. Thus, it is critical to understand the effect of noise on expectation values in order to develop bounds on both QAOA parameter training and performance guarantees.

The effective dynamics of an expectation value are captured by a cumulant expansion of the toggling-frame error propagator [37,38]. Using $\rho(t) = \tilde{U}_E(t)U_0(t)\rho(0)U_0^\dagger(t)\tilde{U}_E^\dagger(t)$, the noise-averaged, time-dependent expectation value can be rewritten as

$$\overline{\langle O(T) \rangle} = \overline{\mathrm{Tr}[\tilde{U}_E(T)\rho_0(T)\tilde{U}_E^\dagger(T)O]} = \mathrm{Tr}[\Lambda(T)\rho_0(T)O]. \tag{21}$$

The reason for the particular rotating-frame choice is now evident: it allows for the dynamics to be partitioned into an error operator $\Lambda(T) = \overline{O^{-1}\tilde{U}_E^\dagger(T)O\tilde{U}_E(T)}$ and the ideal time-evolved state $\rho_0(T)$. The error operator encapsulates all of the error dynamics, i.e., all dynamics generated by the error Hamiltonian. Note that it is observable dependent, but not state dependent. Equation (21) expresses the error dynamics in an intuitive way, where an absence of $H_E(t)$ results in $\Lambda(T) = I$. This perspective can be quite useful, especially when one seeks to mitigate the effects of the noise and, thus, minimize the distance between $\Lambda(T)$ and the identity operator. It is important to note that $\Lambda(T)$ demands O to be invertible; however, as we discuss in Appendix D 1, if one can write a general O as a sum of invertible operators then a similar expression to Eq. (21) can be obtained.

In the case where O is invertible, the error operator can be expressed as a cumulant expansion such that

$$\overline{\langle O(T)\rangle} = \mathrm{Tr}[e^{\mathcal{C}(T)}\rho_0(T)H_C]. \tag{22}$$

$\mathcal{C}(T) = \sum_{k=1}^{\infty}(-i)^k\mathcal{C}_k(T)/k!$ denotes the cumulant series with $\mathcal{C}_k(T)$ being the kth cumulant [49]. Assuming a weak-noise regime, i.e., the dynamics are dominated by the control, we truncate the series at second order: $\mathcal{C}(T) \approx -i\mathcal{C}_1(T) - \mathcal{C}_2(T)/2$. See Appendix C for more details on the cumulant expansion.

C. Constant coherent errors

Constant coherent errors are trivially captured by the cumulant expansion, most notably when the error mean is small, i.e., $\eta_\mu \ll 1$. In this case, the dynamics are well characterized by the first cumulant:

$$\mathcal{C}_1(T) = \sum_{\mu=M,C}\eta_\mu\int_0^T dt(\tilde{H}_\mu(t) - O^{-1}\tilde{H}_\mu(t)O).$$

The QAOA evolution is piecewise constant which allows for the dynamics to be partitioned into a sum of integrals in which the control propagator can be expressed as

$$U_0(T,t) = \begin{cases} Q_{p:j+1}e^{-i(t_{2j}-t)H_M} & : \quad t \in \Delta t_j^{\mathrm{M}} \\ Q_{p:j}U_C^\dagger(\gamma_j)e^{-i(t_{2j-1}-t)H_C} & : \quad t \in \Delta t_j^{\mathrm{C}}, \end{cases} \tag{23}$$

where

$$Q_{k:j} = U_M(\beta_k)U_C(\gamma_k)\cdots U_M(\beta_j)U_C(\gamma_j). \tag{24}$$

Examining the error Hamiltonian and the control propagator, one finds that certain commutations naturally arise between terms that comprise $H_E(t)$ and $U_C(T,t)$. As a result, the first cumulant reduces to integrals over constants and, thus,

$$\mathcal{C}_1(T) = \eta_M\sum_{j=1}^{p}\beta_j\mathcal{H}_{M,j+1}^{(1)} + \eta_C\sum_{j=1}^{p}\gamma_j\mathcal{H}_{C,j}^{(1)}, \tag{25}$$

where the operator $\mathcal{H}_{\mu,j}^{(1)} = \mathcal{Q}_j(H_\mu) - O^{-1}\mathcal{Q}_j(H_\mu)O$ with $\mu = M, C$. Note the inclusion of the superoperator $\mathcal{Q}_j(H) = Q_{p:j}HQ_{p:j}^\dagger$ that results from the rotating frame.

D. Stochastic errors

Following a similar procedure, one can compute the cumulant expressions for the stochastic precision error model. The zero-mean assumption leads to $\mathcal{C}_1(T) \equiv 0$ and, thus, the dominant term is the second cumulant. There are a number of features of our noise model that allow $\mathcal{C}_2(T)$ to be conveniently simplified.

(1) The statistical properties of the noise, namely, the lack of cross-correlations, allows for the second cumulant to be partitioned into terms solely proportional to $\tilde{H}_M(t)$ or $\tilde{H}_C(t)$, i.e., $\mathcal{C}_2(T) = \mathcal{C}_{2,M}(T) + \mathcal{C}_{2,C}(T)$.

(2) When combined with the piecewise-constant nature of the error Hamiltonian, the cumulant expressions reduce to a sum of nested integrals only over the domain in which $t_{j-1} \leqslant s_2 \leqslant s_2 \leqslant t_j$.

(3) Leveraging the definition of the ideal QAOA evolution given in Eq. (13) and exploiting commutations between the error Hamiltonian and the ideal propagator, the above expressions become integrals over constants.

Together, these features result in

$$\mathcal{C}_2(T) = \frac{\Gamma_M}{2}\sum_{j=1}^{p}\beta_j\mathcal{H}_{M,j+1}^{(2)} + \frac{\Gamma_C}{2}\sum_{j=1}^{p}\gamma_j\mathcal{H}_{C,j}^{(2)}, \tag{26}$$

where the operator $\mathcal{H}_{\mu,j}^{(2)} = \mathcal{Q}_j(H_\mu^2) + O^{-1}\mathcal{Q}_j(H_\mu^2)O - 2O^{-1}\mathcal{Q}_j(H_\mu)O\mathcal{Q}_j(H_\mu)$ with $\mu = M, C$.

It is important to note that, despite the equivalence between stochastic (uncorrelated) unitary control errors and Markovian channels, the cumulant approach affords many advantages over standard completely positive trace-preserving (CPTP) map formulations. Importantly, the cumulant approach enables investigations of a wide range of correlated errors and, thus, it is not limited to static or uncorrelated noise [37–41]. We focus on the latter here to make a direct comparison with results from QA and leave a broader study of correlated precision errors for future studies. Furthermore, the cumulant approach results in concise and intuitive expressions for the error dynamics [Eqs. (25) and (26)] that are exceptionally convenient for deriving bounds on performance guarantees and training error, as we show in the next section.

V. ERROR BOUNDS

Perturbative expansions, like Dyson and Magnus, have proven to be useful tools for developing bounds on open quantum system dynamics [50]. Such bounds have been utilized to investigate and evaluate control schemes designed to mitigate unwanted environmental interactions [51–53]. Naturally, the cumulant expansion employed here affords a similar capability that we will exploit to bound various metrics relevant to QAOA.

A. Expectation value

Here, we provide bounds on the absolute error and MSE between the ideal and faulty expectation value of an observable O. We define the ideal average energy $\langle O(T)\rangle_0$ as the expectation value resulting from the system being time evolved with respect to $U_0(T)$. Similarly, the faulty average energy $\overline{\langle O(T)\rangle}$ arises from the time evolution of $U(T)$. In the former, the error dynamics are equivalent to the identity operator, i.e., $\Lambda(T) \equiv I$.

The absolute error is given by

$$\begin{aligned} |\overline{\Delta\langle O(T)\rangle}| &= |\overline{\langle O(T)\rangle} - \langle O(T)\rangle_0| \\ &= |\mathrm{Tr}[(\Lambda(T) - I)\rho_0(T)O]|. \end{aligned} \tag{27}$$

Utilizing the trace inequality $|\mathrm{Tr}(AB)| \leqslant \|A\|_\infty \|B^\dagger\|_1$, where $\|\cdot\|_\infty$ is the largest singular value and $\|\cdot\|_1$ is the sum of singular values $s_i(B)$ [50], we isolate the norm of the error dynamics $\|\Lambda(T)\|_\infty$ from that of error-free operators $\|O\rho_0(T)\|_1$. Using the cumulant expansion to approximate $\Lambda(T)$ and employing submultiplicativity and the triangle inequality, we arrive at the upper bound

$$|\overline{\Delta \langle O(T)\rangle}| \leqslant (e^{\|\mathcal{C}(T)\|_\infty} - 1)\|O\rho_0(T)\|_1. \tag{28}$$

Note that if $\|\mathcal{C}(T)\|_\infty \leqslant 1$, then the inequality $e^x - 1 \leqslant (e-1)x$ leads to $|\overline{\Delta \langle O(T)\rangle}| \leqslant \|\mathcal{C}(T)\|_\infty \|O\rho_0(T)\|_1$. In both cases, the error between the ideal and faulty expectation values is dependent upon the distance between $\|\mathcal{C}(T)\|_\infty$ and zero.

In addition to the absolute error, we provide bounds on the MSE. Representing the second moment in the error, the MSE measures the quality of an estimator in terms of both the variance and bias. We define the estimator as $\langle O(T)\rangle$ and the MSE as

$$\begin{aligned}\overline{\mathrm{MSE}(O(T))} &= \overline{\langle (O(T) - \langle O(T)\rangle_0)^2\rangle} \\ &= \overline{\mathrm{Var}(O(T))} + (\overline{\Delta \langle O(T)\rangle})^2,\end{aligned} \tag{29}$$

where $\mathrm{Var}(O(T)) = \langle O^2(T)\rangle - \langle O(T)\rangle^2$ is the variance in the faulty observable expectation value. Note that defining the estimator in this particular way will result in a bound on the MSE in the asymptotic regime when the number of samples $N_S \to \infty$. Finite sampling effects are further discussed in Appendix D.

We derive two bounds on the MSE: one specific to constant coherent errors and another, more general, bound characterized by the operator norm of $\Lambda(T)$. The former utilizes formal integration of $\tilde{U}_E(t)$—similar to typical Trotter error bounding—to upper bound the MSE in terms of $\|H_E(t)\|_\infty$. The latter employs the techniques used in Eq. (28) and applies to any spatiotemporally correlated, weak-noise model. The second approach inherently provides a bound for constant coherent errors as well; however, we find that the first approach yields a tighter bound than its more general counterpart for this particular error model.

In the case of coherent errors, the MSE is bounded as

$$\mathrm{MSE}(O(T)) \leqslant 2\|O\|_\infty^2 \big(1 + 2h_E^2\big), \tag{30}$$

where

$$h_E = \theta_M(T)\|H_M\|_\infty + \theta_C(T)\|H_C\|_\infty \tag{31}$$

with $\theta_M(T) = \int_0^T dt|\eta_M(t)|[1 - s(t)]$ and $\theta_C(T) = \int_0^T dt\, |\eta_C(t)|s(t)$. This result follows from a bound on the variance $\mathrm{Var}(O(T)) \leqslant 2\|O\|_\infty^2$ and on the bias $\Delta\langle O(T)\rangle) \leqslant \int_0^T \|H_E(t)\|_\infty dt \leqslant h_E$. Further details on the derivation can be found in Appendix E.

More generally, the variance can be bounded in terms of the error operator according to

$$\begin{aligned}\overline{\mathrm{Var}(O(T))} \leqslant \|\Lambda(T)\|_\infty \big(&\|\Lambda(T)\|_\infty \|O\rho_0(T)\|_1^2 \\ &+ \|O^2\rho_0(T)\|_1\big).\end{aligned} \tag{32}$$

This follows directly from an application of the triangle inequality and the variant of Hölder's inequality used in Eq. (28). A bound on the bias [the second term in Eq. (29)] is obtained in a similar manner to that of Eq. (28), namely,

$$(\overline{\Delta \langle O(T)\rangle})^2 \leqslant \|\Lambda(T) - 1\|_\infty^2 \|O\rho_0(T)\|_1^2. \tag{33}$$

The MSE achieves a quadratic scaling in the norm of the error operator and, therefore, the cumulant sum. This can be observed by explicitly expressing the MSE in terms of the norm of the cumulant sum. If $\|\mathcal{C}(T)\|_\infty \leqslant 1$ is assumed, the MSE scales as $\mathcal{O}(\|\mathcal{C}(T)\|_\infty^2)$, explicitly conveying a quadratic scaling.

B. Approximation ratio

The bound derived in Eq. (28) naturally leads to a bound on the approximation ratio. Specifically, we aim to bound the difference between the faulty and ideal approximation ratios for evolution dictated by the optimal *ideal* variational parameters. Above, we showed that the absolute error can be expressed in terms of the operator norm of the cumulant sum $\mathcal{C}(T)$. As a result, one may utilize this bound to quantify the effect of various types of precision errors on the approximation ratio. While we focus on a general bound here, subsequent sections consider specific canonical QAOA problems.

In the ideal scenario, an ϵ^*-approximate QAOA achieves a performance guarantee with the variational parameters $(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)$, or equivalently a total algorithmic run-time T^*. The optimal approximation ratio $\epsilon^* = F(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)/C_{\max}$ is expressed in terms of the average energy with respect to the problem Hamiltonian, $F(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)$, i.e., the typical QAOA objective function.

In order to evaluate the error induced by $H_E(t)$, and more specifically precision errors, consider the average energy resulting from the faulty dynamics $\widetilde{F}(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)$. As can be anticipated, the average energy can be associated with a faulty approximation ratio $\widetilde{\epsilon^*} = \widetilde{F}(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)/C_{\max}$. The upper bound on the absolute error $|\Delta\epsilon^*| = |\widetilde{\epsilon^*} - \epsilon^*|$ is

$$\begin{aligned}|\overline{\Delta\epsilon^*}| &= \frac{1}{C_{\max}}|\widetilde{F}(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*) - F(\boldsymbol{\gamma}^*, \boldsymbol{\beta}^*)| \\ &\leqslant \frac{1}{C_{\max}}(e^{\|\mathcal{C}(T^*)\|_\infty} - 1)\|H_C\,\rho_0(T^*)\|_1,\end{aligned} \tag{34}$$

which follows directly from Eq. (28).

C. Variational parameter training

QAOA variational parameter training routinely involves a classical gradient descent algorithm. Gradients are either analytically determined or approximated via a finite difference approximation. In either case, parameters are updated based on estimates of the average energy using states time evolved by the QAOA with variational parameters acquired from previous iterations. As such, when presented with a faulty QAOA, subject to systematic or environment noise sources, errors induced in the expectation value propagate to the gradient and, ultimately, impact parameter training.

1. Open-loop training

Here, we aim to evaluate the impact of errors on the gradient by bounding the absolute error in the gradient of the expectation value. First, we focus on open-loop (offline) optimization protocols that utilize exact gradient expressions for parameter training. We define the error between the faulty and ideal gradients for a single variational parameter:

$$|\partial_{\gamma_j}\overline{\Delta\langle O(T)\rangle}| = |\partial_{\gamma_j}\overline{\langle O(T)\rangle} - \partial_{\gamma_j}\langle O(T)\rangle_0|. \tag{35}$$

Applying the triangle inequality, the error can be separated into two terms as follows:

$$\left|\partial_{\gamma_j}\overline{\Delta\langle O(T)\rangle}\right| \leqslant |\mathrm{Tr}[(\partial_{\gamma_j}\Lambda(T))\rho_0(T)O]| + |\mathrm{Tr}[(\Lambda(T)-1)\partial_{\gamma_j}\rho_0(T)O]|. \tag{36}$$

The first term is characterized by the differentiation of the error operator, while the latter includes a differentiation of the ideal time-evolved density operator. We bound each term individually, starting with

$$\left|\mathrm{Tr}\big[\big(\partial_{\gamma_j}\Lambda(T)\big)\rho_0(T)O\big]\right| \leqslant \|\partial_{\gamma_j}\Lambda(T)\|_\infty\ \|O\rho_0(T)\|_1 \leqslant \lambda(T)\ \|O\rho_0(T)\|_1. \tag{37}$$

The first inequality follows from an application of Hölder's inequality, while the second incorporates

$$\left\|\partial_{\gamma_j}\Lambda(T)\right\|_\infty \leqslant e^{\|\mathcal{C}(T)\|_\infty}\frac{e^{2\|\mathcal{C}(T)\|_\infty}-1}{2\|\mathcal{C}(T)\|_\infty}\left\|\partial_{\gamma_j}\mathcal{C}(T)\right\|_\infty = \lambda(T). \tag{38}$$

Briefly, this bound is obtained by expressing $\Lambda(T)$ as an exponential of the cumulant series and using the general definition of a derivative of an exponentiated operator.

The second term in Eq. (36) can be bounded via Hölder's inequality and further simplified using the Liouville–von Neumann equation. Namely, we find

$$\left|\mathrm{Tr}\big[(\Lambda(T)-1)\partial_{\gamma_j}\rho_0(T)O\big]\right| \leqslant \|\Lambda(T)-1\|_\infty\|\tilde{O}(T_j,T)[O,\rho_0(T_j)]\|_1, \tag{39}$$

where $\tilde{O}(T_j,T) = Q_{p:j}OQ_{p:j}^\dagger$. Note that the second term in Eq. (39) follows from observing the connection between $\|O\partial_{\gamma_j}\rho_0(T)\|_1$ and the dynamical equation for the density matrix up to time T_j, i.e., $\partial_{\gamma_j}\rho_0(T) = -iQ_{p:j}[H_C,\rho_0(T_j)]Q_{p:j}^\dagger$. Note that both terms scale exponentially in the cumulant and, thus, suffer the same susceptibility to precision errors as the approximation ratio. Appendix E 3 contains further elaboration on this bound and the others shown above.

2. Closed-loop training

Commonly, QAOA training leverages approximate gradient expressions in place of exact gradients. This most notably occurs when closed-loop optimization protocols employ some degree of hardware-in-the-loop querying to train variational parameters. First-order gradients are calculated via finite difference approximations and subsequently incorporated into stochastic gradient descent protocols, such as simultaneous perturbation stochastic approximation (SPSA) [54].

The training error in stochastic optimization protocols can be associated with the quality of the estimator and, thus, the MSE. In the asymptotic limit, this training error can be bounded by the expressions given in Sec. V A. Coherent errors yield an upper bound of Eq. (30) on the training error, while the sum of Eqs. (32) and (33) specifies a general bound in terms of the error operator. Finite sampling effects are pertinent to approximate gradient-based algorithms. Contributing additional corrections to the above bounds, such effects are elaborated upon in Appendix D 4.

D. Observable-independent measures

Error measures expressed in terms of expectation values provide insight into the deleterious effects of noisy QAOA evolution with respect to the measurement of an observable. Although such measures are of practical importance for QAOA, from a theoretical perspective it can be constructive to eliminate state and observable dependence and focus solely on the unitary propagator. Such distance measures are commonly used in optimized control [55] and Trotter error analysis [56] to achieve a holistic understanding of error propagation.

In an effort to obtain such an understanding for the precision error problem in QAOA, we examine the distance between the faulty QAOA and ideal QAOA evolution:

$$\Delta U = \|U(T) - U_0(T)\|_2. \tag{40}$$

Note that $\|A\|_2 = \sqrt{\mathrm{Tr}A^\dagger A}$ defines the Frobenius norm. We bound this quantity, focusing particularly on the ensemble-average-squared distance, i.e.,

$$\overline{(\Delta U)^2} = \mathrm{Tr}[2I - \overline{\tilde{U}_E(T)} - \overline{\tilde{U}_E^\dagger(T)}]. \tag{41}$$

The above expression includes a noise-averaged unitary that differs from that of the error dynamics introduced in Eq. (22). Namely, the unitary is not generated by the effective Hamiltonian, but rather $\tilde{H}_E(t)$. This does not create any issues, however, as a cumulant expansion $\overline{\tilde{U}_E(T)} = \overline{\mathcal{T}_+ e^{-i\int_0^T \tilde{H}_E(t)dt}} = e^{\mathcal{C}_U(T)}$ can still be obtained, with $\mathcal{C}_U(T)$ denoting the cumulant terms rewritten in terms of $\tilde{H}_E(t)$. Incorporating the expansion into Eq. (40) leads to

$$\overline{(\Delta U)^2} = \mathrm{Tr}[2I - e^{\mathcal{C}_U(T)} - e^{\mathcal{C}_U^\dagger(T)}] \leqslant 2(e^{\|\mathcal{C}_U(T)\|_\infty} - 1). \tag{42}$$

This bound possesses similar qualities to the bound obtained for the absolute error in expectation values, with the primary distinction being the presence of an *observable-independent* cumulant expansion.

VI. NUMERICAL SIMULATIONS AND EMPIRICAL BOUNDS

In this section, we discuss the effect of precision errors on two specific examples of QAOA circuit implementations. In the first example, we examine the Grover search algorithm using QAOA [5]. Second, we consider obtaining the ground state of the nearest-neighbor Ising model on a ring [43], which also can be mapped to the 2-SAT problem on a ring [57]. While the Grover algorithm provides an example of a QAOA with an analytical solution for the optimal circuit depth and angles, the circuit depth increases exponentially with the number of qubits. In the second example, we obtain the ground state of the Ising problem using a number of QAOA layers that scales linearly in the number of qubits.

We numerically simulate precision errors and investigate their effect on the accuracy of the implementation of the

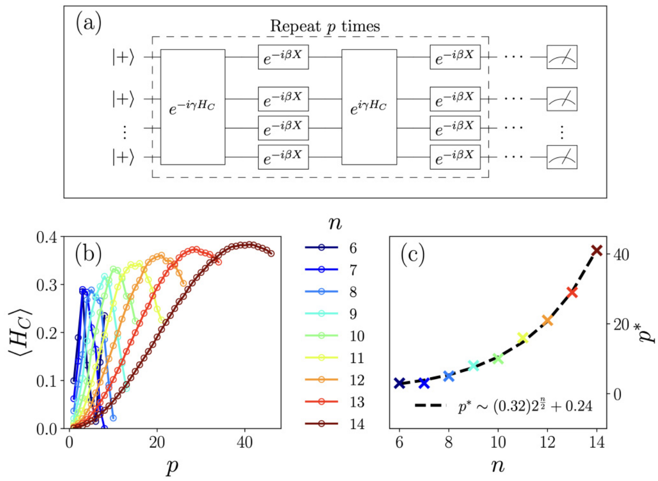

FIG. 2. Noisless implementation of the Grover search algorithm. (a) The circuit for implementing the Grover search algorithm using QAOA [5]. (b) The expectation value of the cost function as a function of the number of layers, p. (c) The optimal number of layers, $p^* \sim 2^{n/2}$, that maximizes the cost function increases proportionally with the problem size. We extract the functional form of the optimal number of layers, $p^* \approx (0.32)2^{n/2} + 0.24$.

algorithm. The errors $\eta_{C,M}$ modify the QAOA phases, $\gamma \rightarrow \gamma(1+\eta_C)$ and $\beta \rightarrow \beta(1+\eta_M)$ [see Eq. (18) for definitions of η_ν]. These errors are drawn from a normal distribution: $\eta_\nu \in \mathcal{N}(\eta, \sqrt{\Gamma})$ with η being the mean, Γ the variance of the normal distribution, and $\nu \equiv M, C$. The stochastic error case reduces to $\eta_\nu \in \mathcal{N}(0, \sqrt{\Gamma})$ [$\Gamma_\nu(t-t') = \Gamma\delta(t-t')$, i.e., no correlation between subsequent errors in γ_k and β_k] and the coherent error case reduces to $\eta_\nu \in \mathcal{N}(\eta, 0)$ [$\Gamma_\nu(t-t') = \eta^2$, i.e., constant errors in all γ_k and β_k]. We average our results over 1000 noise realizations for the case of the stochastic error, and 1 realization for the coherent case.

In order to numerically extract the loss in accuracy of the computation with increasing precision error, we model the noise-averaged expectation value using a decay function dependent on the number of layers and noise parameters,

$$\overline{\langle H_C\rangle} \approx \langle H_C\rangle_0\, \mathcal{S}(p, \eta, \Gamma), \tag{43}$$

where $\langle H_C\rangle_0$ corresponds to the noiseless expectation value. The analytical form of the decay function, $\mathcal{S}(p)$, depends on the nature of the precision errors (stochastic vs coherent) and the number of layers. In the following, we show that the expectation value decays exponentially away from the noiseless value, and that this behavior is generic.

A. Grover's search algorithm

The Grover search problem involves finding a marked element $|m\rangle$ in a unsorted database consisting of $N = 2^n$ elements, where n is the number of qubits required to represent the database. For a marked state of $|m\rangle = |0\cdots 0\rangle$, we have the phase Hamiltonian,

$$H_C = |0\cdots 0\rangle\langle 0\cdots 0| = \bigotimes_{i=1}^{n} \frac{I+Z_i}{2}, \tag{44}$$

and mixer Hamiltonian given in Eq. (9). The phase Hamiltonian corresponds to a fully connected Ising Hamiltonian, with all possible tensor product of Z_i operators. However, note that $||H_C|| = 1$. Also, note that the Hamiltonian encoding the cost function, H_C, while clearly Hermitian, is noninvertible.

Typically, for the Grover search problem, each layer of the QAOA *Ansatz* has the following structure [5]:

$$\begin{aligned} |\psi_{\text{out}}\rangle &= [W(\gamma,\beta)]^p\, |+\rangle^{\otimes n} \\ &= \prod_{k=1}^{2p} (e^{-i\beta H_M} e^{(-1)^{k+1} i\gamma H_C})\, |+\rangle^{\otimes n} \end{aligned} \tag{45}$$

with $\gamma = \pi$ and $\beta = \frac{\pi}{n}$. The QAOA applies the unitary $W(\gamma,\beta)$ p times. The quantum circuit for a single layer is shown in Fig. 2(a). We simulate the Grover QAOA circuits using the built-in simulator in cirq [58]. The expectation value of the cost function, which corresponds to the probability of measuring the marked state, oscillates as a function of the number of layers, p. In Fig. 2(b) we show this probability as a function of p. We obtain an expression for the optimal number of layers for Grover search as a function of n by numerically fitting the first maximum in $\langle H_C\rangle$,

$$p^*(n) \approx a_1 2^{n/2} + a_2, \tag{46}$$

$a_1 = 0.32$, $a_2 = 0.24$ [see Fig. 2(b)]. This scaling of the optimal number of layers for Grover is in line with the theoretical analysis in Ref. [5], namely, $p^* \sim O(\sqrt{N})$.

1. Empirical bounds on average energy

We begin by investigating the effect of precision errors on Grover's search from a numerical perspective, and develop empirical bounds to which we compare our analytical results above. Note the Grover cost function, H_C, is not invertible, which poses an issue for the cumulant expansion and the definition of the error operator $\Lambda(T)$ [Eq. (21)]. However, we can be rewrite H_C as

$$\begin{aligned} H_C &= \frac{1}{N} I + O, \\ \text{where } O &= \frac{1}{N}\sum_{\{A_i\}} A_1 \otimes \cdots \otimes A_n \end{aligned} \tag{47}$$

where $A_i \in \{I, Z_i\}$ and O is invertible and traceless. Using this representation of H_C, we can employ the bounds derived in the previous sections to bound the expectation value of H_C. However, it will also assist in empirical investigations of the average energy. In particular, we expect a noise-averaged expectation value of

$$\begin{aligned} \overline{\langle H_C\rangle} &= \frac{1}{N} I + \overline{\langle O\rangle} = \frac{1}{N} I + \langle O\rangle_0\, \mathcal{S}_O(p^*) \\ &= \frac{1}{N}[1 - \mathcal{S}_O(p^*)] + \langle H_C\rangle_0\, \mathcal{S}_O(p^*), \end{aligned} \tag{48}$$

where $\mathcal{S}_O(p^*)$ is the decay function specific to the operator O.

In Fig. 3, we numerically study the effect of precision error for a range of noise strengths. Figures 3(a) and 3(b) display $\overline{\langle H_C\rangle}$ as a function of p for stochastic and coherent errors, respectively. Focusing on the optimal QAOA order

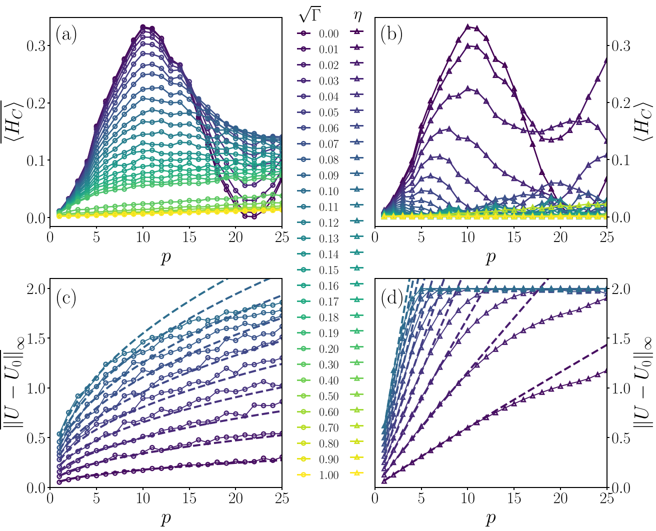

FIG. 3. Performance of Grover QAOA circuits as a function of number of layers, p, in the presence of precision errors. We consider the effect of two types of precision errors: (i) stochastic errors, drawn from a normal distribution $\mathcal{N}(0, \sqrt{\Gamma})$ (left column), and (ii) coherent errors, drawn from a normal distribution $\mathcal{N}(\eta, 0)$ (right column). The top row shows the effect of the precision errors on the calculated expectation value, $\overline{\langle H_C \rangle}$, where $\overline{\cdots}$ indicates noise averaging. The bottom row shows the average distance between the faulty and the perfect QAOA unitary as characterized by the average difference in the ∞-norm, $\overline{\|U - U_0\|_\infty}$. The dashed line shows the behavior of a scaling function that describes the behavior of the increasing separation for stochastic and coherent errors; the numerical parameters are shown in Appendix F 1. The simulations are done for $n = 10$ qubits, with 1000 noise realizations for the stochastic case and 1 realization for the coherent error case.

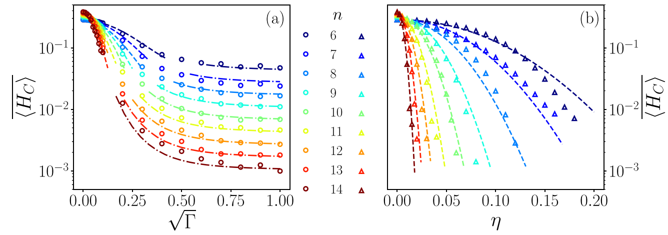

FIG. 4. Performance of Grover QAOA for the optimal number of layers (p^*) as a function of noise strength [(a) Γ or (b) η] and system size (color). (a) Noise-averaged expectation value, $\overline{\langle H_C \rangle}$, as a function of the strength of the stochastic error (open circles). The lines (dashed and dash-dotted) indicate the behavior from the scaling function. Dashed lines denote fits using Eq. (49), while dash-dotted lines correspond to Eq. (F4) discussed in Appendix F. (b) Expectation value as a function of strength of coherent error (open triangles) and the corresponding scaling function (dashed line).

p^*, we develop a phenomenological fitting function that captures the decay of the expectation value as a function of the noise strength and system size. We identify two regimes: the weak-noise limit, where $\Gamma p^* \ll 1$ and $\eta p^* \ll 1$, and the strong-noise limit, where both quantities are greater than unity. In the former, the noise-averaged expectation value can be approximated as $\overline{\langle H_C \rangle} \approx \langle H_C \rangle_0 \mathcal{S}_O(p^*)$.

A comparison of $\overline{\langle H_C \rangle}$ as a function of noise strength for the optimal p is shown in Fig. 4. In the weak-noise limit, the decay function takes the form $\mathcal{S}_O(p^*) \approx \exp[-\xi_O(p^*)]$ with

$$\xi_O(p^*) = \begin{cases} -\Gamma p^*/\Gamma_s, & \text{[stochastic error]} \\ -\eta^2 p^{*2}/\eta_c^2, & \text{[coherent error]}, \end{cases} \tag{49}$$

where $\Gamma_s \approx 0.22$ and $\eta_c \approx 3.46$ are decay constants. While Fig. 4(b) only displays the weak-noise limit, Fig. 4(a) displays both limits. The weak-noise decay function fit is given by the dashed line, while the dash-dotted line denotes the strong-noise limit. The fitting function for the strong-noise limit is described in detail in Appendix F.

The calculated expectation value decays exponentially with precision error scale. Note that the implementation of the optimal Grover algorithm requires p^* to grow with problem size. Conversely, our results imply that for an accurate implementation of the optimal Grover QAOA circuit, the precision errors need to be bounded from above: $\Gamma \ll \Gamma_s/p^*$ and $\eta \ll \eta_c/p^*$.

2. Comparison against analytical bounds

We compare the numerically obtained $\mathcal{S}_O(p^*)$ to the decay functions derived via the cumulant expansion and its associated bounds. Specifically, we focus on the absolute error in the expectation value, and compare the numerically obtained absolute error to three different cumulant calculations for the case of stochastic precision errors. First, the absolute error is estimated by approximating the expectation value dynamics using the second-cumulant-truncated error operator. The cost Hamiltonian is expressed as a sum of two terms in Eq. (47); however, it can also be written as a sum of N terms: $O = 1/N \sum_i O_i$, where $O_i \in \boldsymbol{O}$ correspond to the summands of Eq. (47). Using the latter representation, the expectation value takes the form

$$\overline{\langle H_C \rangle} = \frac{1}{N} \sum_{\mathcal{X} \in \{I, \boldsymbol{O}\}} \mathrm{Tr}[\Lambda_{\mathcal{X}}(T) \rho_S(T) \mathcal{X}]. \tag{50}$$

By truncating each error operator as $\Lambda_{\mathcal{X}}(T) \approx e^{-\mathcal{C}_2^{\mathcal{X}}(T)/2}$, we obtain an expression for the approximate dynamics of the expectation value of H_C. Subsequently, an estimate of the absolute error is obtained by substituting Eq. (50) into Eq. (27).

The approximate dynamics are compared against bounds on the absolute error derived in Sec. V A. Specifically, we consider a variant of the upper bound given in Eq. (28) for noninvertible observables:

$$|\overline{\Delta \langle H_C \rangle}| \leqslant \frac{1}{N} \Big\| \sum_{\mathcal{X} \in \{I, \boldsymbol{O}\}} \mathcal{X} \Lambda_{\mathcal{X}}(T) - H_C \Big\|_\infty \tag{51}$$

(see Appendix D 1 for further details). We consider a numerical evaluation of this upper bound by truncating the error operator to second order and calculating the bound numerically. In addition, we estimate the bound analytically via

$$\begin{aligned} |\overline{\Delta \langle H_C \rangle}| &\leqslant \frac{1}{N} \sum_{\mathcal{X} \in \{I, \boldsymbol{O}\}} \|\mathcal{X}\|_\infty \big(e^{\|\mathcal{C}_2^{\mathcal{X}}(T)\|_\infty/2} - 1\big) \\ &\leqslant \min[(e^{4\pi p^* \Gamma} - 1), \|H_C\|_\infty], \end{aligned} \tag{52}$$

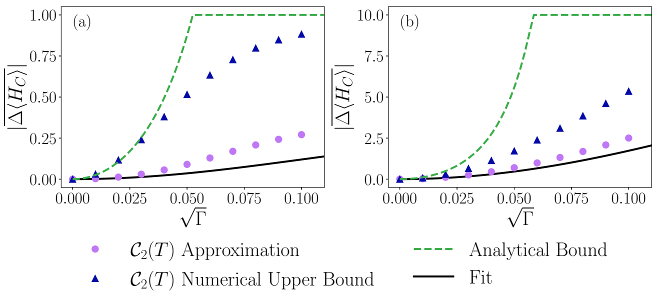

FIG. 5. Comparison of the numerically obtained scaling functions, with the cumulant expression and bounds for stochastic precision errors. Second-cumulant approximation (purple circles) shows good agreement with scaling functions (black lines) in the weak-noise regime. Numerically calculated error bounds using the second-cumulant approximation (blue triangles) are shown in addition to analytically calculated cumulant bounds (green dashed) of Eq. (28). Plots summarize results for $n = 10$ qubits for (a) Grover's search and (b) GHZ state preparation.

where the upper bound on the second cumulant is

$$\left\|\mathcal{C}_2^{\mathcal{X}}(T)\right\|_\infty \leqslant 8\pi p^*\Gamma \sim 8\pi\sqrt{N}\Gamma. \tag{53}$$

The bound on the $\|\mathcal{C}_2^{\mathcal{X}}(T)\|_\infty$ holds for each constituent observable $\mathcal{X}$ defined for Grover's search. Note that the upper bound expression is clipped at a maximum value of $\|H_C\|_\infty$ as this represents the physical bound on possible QAOA solutions.

The results of the cumulant comparison are summarized in Fig. 5(a) for Grover's search. The second-cumulant approximation (purple circles) agrees well the numerically obtained fit in the weak-noise regime. Analytical bounds (green dashed lines) provide a modest representation of precision error dependence, deviating more substantially as the variance increases. Numerical bounds (blue triangles) and approximate dynamics follow similar trends, indicating that, as the variance increases, higher-order cumulants likely contribute more significantly to the dynamics.

3. Bounds on distance between unitaries

In addition, we evaluate the effect of precision errors by calculating the average distance between the faulty and ideal unitary. Our results are summarized in Figs. 3(c) and 3(d) for stochastic and coherent error cases, respectively. In the limit of weak noise, the average distance

$$\overline{\|U - U_0\|_\infty} \propto \begin{cases} \sqrt{\Gamma p}, & \text{[stochastic error]} \\ \eta p, & \text{[coherent error]}. \end{cases} \tag{54}$$

Hence, the average distance between the noisy and noiseless unitary grows with the number of layers of the QAOA circuit.

Finally, we consider the limit of large stochastic noise, $\Gamma p^* \gg 1$. Assuming that the output of a noisy QAOA evolution is a completely random state, one would expect that $\overline{\langle H_C\rangle} \sim 1/N = 1/2^{n/2}$. However, in this limit, the noisy QAOA does not give rise to uniformly random states. Instead, we observe saturation in the expectation value that scales with problem size according to

$$\langle H\rangle_{\text{sat}} \propto 2^{\xi n}, \tag{55}$$

where $\xi \approx 0.67$. The approach to the saturation value is governed by a power-law behavior,

$$\overline{\langle H_C\rangle} - \langle H_C\rangle_{\text{sat}} \propto \frac{1}{(\Gamma p^*)^\alpha}, \tag{56}$$

where $\alpha \approx 1.61$. The details for the fit to the various parameters are provided in Appendix F.

B. Ising instances

Essentially all combinatorial optimization problems may be cast as two-body Ising Hamiltonians [59,60] of the form $H = \sum_{ij} J_{ij}Z_iZ_j + \sum_i h_iZ_i$. It is therefore natural to study the effects of finite precision errors on problems of this type. A paradigmatic model of the above form, which is therefore convenient to analyze, is the one-dimensional (1D), nearest-neighbor Ising model. The problem Hamiltonian of this model for n qubits is given by

$$H_C = \sum_{i=1}^{n} Z_iZ_{i+1}, \tag{57}$$

where the summation is taken over neighboring spins on a 1D chain with periodic boundary conditions, i.e., $Z_{n+1} \equiv Z_1$. For instance, this particular cost function can be used to find the solutions for 2-SAT on a ring [57].

The cost function Hamiltonian for this Ising model is maximized by a state that is any superposition of all qubits in the $|0\rangle$ or $|1\rangle$ state. In fact, by restricting to a given parity sector satisfying $\prod_{i=1}^n X_i = 1$, this cost function can be used in the QAOA setting as an algorithm for preparing a GHZ state [12,43] $|\text{GHZ}\rangle = 1/\sqrt{2}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n})$. We follow Ref. [43] and use the QAOA *Ansatz* for preparing the GHZ state. Note that we have flipped the sign of the cost Hamiltonian compared with Ref. [43] as we have recast the objective from minimization to maximization. A similar procedure is utilized for the mixer Hamiltonian H_M defined by Eq. (9).

In Ref. [43], it was shown that the optimal number of layers to prepare a GHZ state scales polynomially with the number of qubits. In particular, $p^* = n/2$, with the following structure for the variational circuit *Ansatz*:

$$|\text{GHZ}\rangle = \left(\prod_{k=1}^{n/2} e^{-i\beta_k H_M} e^{-i\gamma_k H_C}\right) |+\rangle^{\otimes n}. \tag{58}$$

The optimal parameters to generate GHZ states for different system sizes are taken from Appendix A of Ref. [43] to test the susceptibility of the state preparation to precision errors.

1. Empirical bounds on average energy

We numerically model the noise-averaged expectation value as a function of the noise strength and determine scaling fits for the GHZ state preparation problem. A summary of the results is shown in Fig. 6 for both stochastic and constant precision errors. Numerical simulations are performed using 1000 realizations for stochastic errors and 1 realization for the coherent error case, using code adapted from Refs. [61,62]. Data collected over various system sizes are utilized to extract an effective scaling of the average expectation value as a function of the noise parameter and optimal QAOA order,

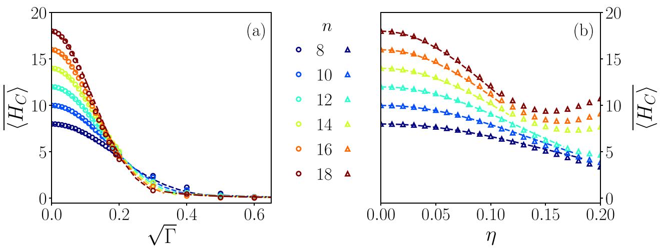

FIG. 6. Preparation of the GHZ state using the QAOA with the optimal number of layers ($p = p^*$) as a function of noise strength [(a) Γ or (b) η] corresponding to noise generated from a normal distribution $\mathcal{N}(\eta, \sqrt{\Gamma})$. (a) Average expectation value (open circles) of the cost function for the case of stochastic error ($\eta = 0$); the lines (dashed and dash-dotted) indicate the scaling fit to the data. (b) Expectation value (open triangles) as a function of the strength of coherent error ($\Gamma = 0$) and the corresponding scaling fit shown as a dashed line.

which is a function of system size. In the weak-noise limit ($\Gamma p^* \ll 1$), the expectation value decays exponentially according to $\langle H_C \rangle \approx \langle H_C \rangle_0 \exp[-\xi(p^*)]$, where

$$\xi(p^*) \approx \begin{cases} -\Gamma p^*/\Gamma_s, & \text{[stochastic error]} \\ -\eta^2 p^*/\eta_c^2, & \text{[coherent error]} \end{cases} \tag{59}$$

with decay constants $\sqrt{\Gamma_s} = 0.51$ and $\eta_c = 0.46$. Note that in the case of large stochastic noise, $\Gamma p^* \gg 1$, we see that the noise-averaged cost function approaches zero in a power law, $\overline{\langle H_C \rangle} \propto 1/(\Gamma p^*)^\alpha$, with $\alpha = 1.46$. Additional information regarding the fitting functions is given in Appendix F.

2. *Comparison against analytical bounds*

The scaling fit is compared against the cumulant expansion and its associated bounds in Fig. 5(b). In general, this comparison will require a similar approach to that used in the Grover search analysis. Namely, we must calculate the approximate dynamics and bounds by summing the contribution of each constituent $Z_i Z_{i+1}$ to the faulty expectation value of H_C. However, for the case considered here, $n = 10$, the inverse of H_C does exist. Thus, no summation is required in the calculation of $\overline{\langle H_C \rangle}$ and the bounding expression given in Eq. (28) directly applies.

Overall, we find good agreement between the second-cumulant dynamics and the fit in the weak-noise regime. Numerically calculated bounds (blue triangles) and second-cumulant dynamics (purple circles) track closely, indicating that the bound is tight for this particular problem class. The analytical bound (green dashed line) is determined by further bounding Eq. (28) via

$$\|\mathcal{C}_2(T)\|_\infty \leqslant n\Gamma\left(1 + \frac{3n}{\sigma_{\min}}\right)T_{\text{GHZ}}, \tag{60}$$

where $\sigma_{\min}$ is the minimum singular value of H_C and $T_{\text{GHZ}} = \sum_{i=1}^{n/2} |\gamma_i| + |\beta_i|$. As expected, this additional bound allows for general behavior of the dynamics to be captured, but does not provide a tight error bound, most notably for increasing variance.

C. Does training combat precision errors?

In the above analysis, we show that precision errors can have a significant impact on QAOA performance. The scenario we consider establishes a baseline in which the noiseless optimal variational parameters are used to highlight the impact of this error source and support our analytical results. Here, we build on this analysis by considering the potential for utilizing variational parameter training to overcome precision errors.

One of the most prominent features of QAOAs (and variational algorithms in general) is that it affords adaptability through the variational parameters. That is, through adjustment or training of the parameters, one has an intrinsic ability to alter the algorithm to potentially improve its performance. Previous studies, however, have shown that parameter training does not always lead to improved performance in noisy QAOA settings [18]. Training QAOAs in the presence of quantum noise, for example, yields nearly identical variational parameter values with reduced QAOA performance commensurate with the noise strength. This behavior is indicative of an overall flattening of the objective function landscape which limits the performance of QAOAs.

Utilizing the QAOA variant of Grover's search, we assess parameter training as an approach for combating coherent and stochastic precision errors. Grover's search provides a convenient framework for this analysis since the canonical QAOA *Ansatz* [Eq. (45)] only contains two parameters. As such, we perform an exhaustive search over the parameter set to determine if there exist new parameters that circumvent precision errors. In this way, we remove the potential variability in performance due to the choice in classical optimizer. Through numerical simulations and the analytical expressions presented in Sec. IV, we argue that coherent errors are correctable, while stochastic errors are not.

In Fig. 7(a), we plot the noiseless average energy landscape for QAOA Grover as a function of γ and β for $n = 10$ qubits at $p = p^*$. We focus on a particular parameter regime surrounding the near-optimal parameter choice, which is marked by the black or white X. We then subject the algorithm to coherent errors of $\eta_M = 0.075$ and $\eta_C = 0.05$, accounting for a 7.5% error on the mixer and a 5% error on the cost Hamiltonian, and stochastic errors with $\Gamma_M = \Gamma_C = 0.075$. Figures 7(b) and 7(c) summarize the energy landscape for coherent and stochastic errors, respectively. In the latter, the expectation values are averaged over 1000 realizations of the noise.

Based on the analysis in Sec. IV C, the coherent error operator is described by a phase evolution [38,41]: $\Lambda(T) \approx e^{-iC_1(T)}$. In general, $C_1(T)$ is not Hermitian [see Eq. (25)], however, Hermiticity can be achieved via an appropriately chosen observable O, such as a Pauli operator, where O^{-1} is well defined and unitary. In such a case, the error operator is unitary. These properties hold for the decomposition of H_C given in Eq. (47) for Grover's search, and the expansion of any H_C in the n-qubit Pauli basis.

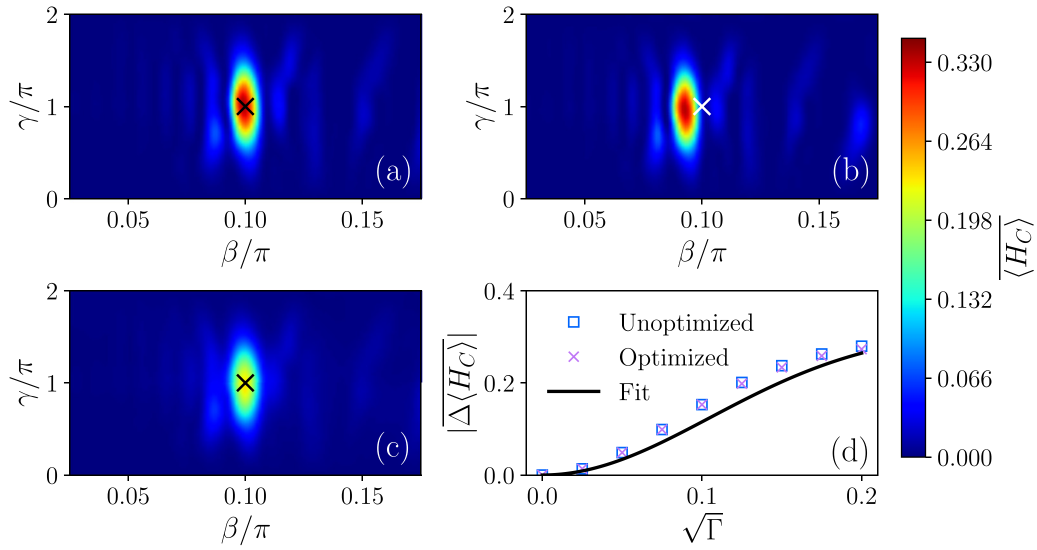

FIG. 7. Optimization of QAOA Grover variational parameters in the presence of precision errors for $n = 10$ qubits. (a) Noiseless objective function landscape as a function of the variational parameters. Noisy objective function landscapes are shown for (b) coherent ($\eta_M = 0.075$, $\eta_C = 0.05$) and (c) stochastic ($\Gamma_M = \Gamma_C = 0.075$) precision errors. Results for stochastic errors are averaged over 1000 realizations of the noise. In panels (a)–(c), the black or white "X" denotes the location of the near-optimal variational parameters discussed in Sec. VI A. The objective function values are captured by the color bar to the right of the plots. (d) The average error in the objective function as a function of the stochastic error variance. The error is plotted for QAOA Grover using the noiseless optimal variational parameters (Unoptimized) and the updated parameters obtained from an exhaustive search (Optimized). The fitting function for Grover's search [Eq. (49)] is included as well for reference and denoted as "Fit." Overall, coherent errors result in a correctable shift in the optimal QAOA solution, while stochastic errors lead to a flattening of the landscape that cannot be corrected by retraining.

Consider a decomposition $H_C = \sum_i c_i O_i$, where c_i are coefficients and O_i is a Pauli operator. The noise-averaged energy can be expressed as

$$\overline{\langle H_C \rangle} \approx \sum_i c_i \mathrm{Tr}\big[e^{-i\mathcal{C}_1^{O_i}(T)} \rho_S(T) O_i\big]$$
$$= \sum_{i,j} c_i \lambda_{ij} \langle \lambda_{ij}(T)|\rho_S(T)|\lambda_{ij}\rangle \,. \qquad (61)$$

The second expression follows from an eigenvalue decomposition $O_i = \sum_j \lambda_{ij} |\lambda_{ij}\rangle\langle\lambda_{ij}|$. In addition, we have defined the rotated state $|\lambda_{ij}(T)\rangle = e^{i\mathcal{C}_1^{O_i}(T)} |\lambda_{ij}\rangle$. From this perspective, the error operator can be viewed as an unwanted rotation that modifies the contribution of the noiseless time-evolved state to the average energy. Figure 7(b) confirms this effect and further illustrates that the coherent error ultimately shifts the location of the maximum energy with respect to the variational parameters. Thus, we refer to this as a correctable error since parameter adjustments can reestablish the maximum energy. This finding is consistent with expectations noted in previous work [63] and experimental demonstrations of variational quantum algorithms [64].

Next, we examine stochastic precision errors. In Fig. 7(c), we show the average energy landscape for QAOA Grover subject to stochastic precision errors. Importantly, the landscape is quite distinct from the coherent error case. We observe a flattening in the average energy landscape, rather than a shift, indicating an overall reduction in the maximum average energy.

We compare the error in the expectation value as a function of noise variance for different variational parameter choices in Fig. 7(d). Blue squares ("Unoptimized") correspond to the error using the noiseless near-optimal variational parameters, while the purple X's ("Optimized") denote the error using the optimal parameters identified from the exhaustive search landscape. The fitting function for Grover's search, given by Eq. (49), is included as a reference and denoted as "Fit." Each data point denotes the average error from 1000 realizations of the noise.

The noisy optimized variational parameters resulting from the exhaustive search are nearly identical to the noiseless parameters, hence the overlapping expectation value error. From this comparison, we conclude that parameter adjustments cannot combat stochastic errors. Our findings are consistent with the analytical expressions in Sec. IV D which specify a decaying error operator $\Lambda(T) \approx e^{-\mathcal{C}_2(T)/2}$ rather than a phase evolution. These results are reminiscent of previous studies of quantum noise in QAOAs [18], and they suggest that stochastic precision errors are as intolerant to parameter training as environmental noise. As we discuss in the next section, fortunately, this does not imply it is impossible to overcome precision errors in QAOA.

VII. DIGITIZATION: AVOIDING PRECISION-INDUCED ERRORS IN QAOA

The analysis in Secs. VI A and VI B conveys that for a p-layer QAOA to achieve a desired accuracy threshold for the expectation value of the cost function, $|\Delta\langle H_C\rangle|/\langle H\rangle_0 = 2^{-\epsilon}$ ($\epsilon > 0$), the maximum precision error permitted in the variational parameters is $\eta_M, \eta_C \propto \sqrt{\epsilon/p}$, where the constant prefactor depends on the details of the problem. While short finite-depth QAOA circuits are currently being investigated for near-term applications in optimization, it is expected that for computationally hard optimization problems it is not unreasonable to expect the number of layers to increase at least superpolynomially with problem size [65–68].

As shown in Sec. VI C, constant coherent over- or under-rotations can be straightforwardly accounted for by updating the optimized angles exactly by the coherent error. However, stochastic errors are not amenable to this approach as they are random in nature. The digitization of the QAOA angles may allow us to avoid errors induced by stochastic precision errors (at the cost of increasing circuit depth).

Consider an N_γ bit implementation of the angle γ in the unitary $\exp(-i\gamma H_C)$, $\gamma \approx \frac{2\pi}{2^{N_\gamma}} \sum_{j=1}^{N_\gamma} A_j 2^j$, where $A_j \in \{0, 1\}$ is the jth bit of the integer closest to $\gamma \frac{2^{N_\gamma}}{2\pi}$. Note that the N_γ bit implementation has a precision error $\sim 2^{-N_\gamma}$. The unitary may be implemented as

$$U_C(\gamma) = \exp(-i\gamma H_C) = \prod_{j=1}^{N_\gamma} \left[\exp\left(i\frac{2\pi}{2^{N_\gamma}} 2^j H_C\right)\right]^{A_j}. \tag{62}$$

A similar analysis can also be extended to the mixer unitary, $U_M(\beta)$. Now, instead of viewing QAOA as an analog quantum algorithm with continuously variable angles, one only needs to implement certain *below-threshold* building block unitaries, $U_{Cj} = \exp(i\frac{2\pi}{2^{N_\gamma}} 2^j H_C)$ and $U_{Mj} = \exp(i\frac{2\pi}{2^{N_\beta}} 2^j H_M)$, with $j = 0, 1 \ldots, N_\gamma$ or N_β. (The term "below-threshold" denotes unitaries that can be implemented with a precision error of at most $\sim\sqrt{\epsilon/pN_{\gamma,\beta}}$.) Together, both unitary operators can be implemented within the desired precision at the cost of increasing the circuit depth $\sim pN_\gamma C_\gamma + pN_\beta C_\beta$, where $C_{\gamma,\beta}$ are the maximum circuit depths necessary to realize one of the unitary operations U_{Cj}, U_{Mj}.

From the above analysis, we conclude that the effects of precision errors larger than $\sqrt{\epsilon/p}$ may be mitigated by an $N_{\gamma,\beta} \propto |\log_2(\sqrt{\epsilon/p})|$ bit implementation of the variational parameters. As a result, mitigating precision errors requires an increase in the circuit depth that is logarithmic with the desired ϵ and p. This would imply that for problems that are hardest for QAOA, where $p \propto 2^n$ (such as Grover's search), one can avoid precision errors at a cost polynomial ($\propto n$) in the number of qubits.

Similar to typical quantum computing gate sets, digitization enables targeted gate design. Compilation strategies need not focus on general compilations of QAOA, but rather highly optimized implementations of $\{U_{C_j}\}_{j=1}^{N_\gamma}$ and $\{U_{M_j}\}_{j=1}^{N_\beta}$. Focused compilation approaches have been shown to be immensely useful in reducing gate depths in quantum tensor network algorithms [69–71]. Furthermore, digitization opens the door to the incorporation of optimal control to achieve high-fidelity constituent QAOA operations that focus on large unitary operations (i.e., partial compilation) as opposed to single- and two-qubit gates [21,72–74]. We leave explorations of various implementations for future studies.

VIII. CONCLUSIONS

In this study, we investigated the effect of precision errors, or the misspecification of variational parameters, on QAOA performance. We provided insight into the extent of their harm on parameter training and performance guarantees. Utilizing concepts from quantum control theory, we analytically estimated the contribution of precision errors to QAOA dynamics. This approach enabled the development of bounds on approximation ratios and parameter training error. Through our analysis, we found that any fixed precision implementation of QAOA will realize performance guarantees that greatly differ from the idealized setting. In particular, we found an exponential reduction in success probability with increasing QAOA order and error magnitude. Numerical studies of the QAOA variant of Grover's search and the one-dimensional transverse-field Ising model conveyed these observations and provided verification for our analytical estimates.

Despite the detrimental nature of precision errors on QAOA performance, we showed that it is possible to mitigate such errors by digitizing the variational parameters. Each constituent QAOA evolution operator was then expanded into a product of operators whose circuit depth is determined by the desired precision. Provided that the decomposition is expressed in terms of evolution operators that can be implemented with a greater precision than that required by the algorithm, one can successfully achieve a desired precision accuracy for QAOA.

While our study focused on precision errors, the analytical approach inherently extends to a far greater class of noise models. Specifically, one can utilize this approach to study the effect of spatial and temporally correlated noise on QAOA and variational algorithms more generally. In summary, we view this framework as providing a unique perspective on variational quantum algorithms and as a constructive tool for predicting faulty algorithm performance and facilitating the development of noise-robust variants.

Following our study, a related work [75] offered an alternative perspective on control errors in QAOA. Its focus is quantifying the effect of such errors on the average energy. Our study provides rigorous bounds on both the approximation ratio and training error, leveraging techniques that can be extended to spatiotemporally correlated noise as well. Furthermore, we offer a compilation scheme for combating control errors.

ACKNOWLEDGMENTS

G.Q., P.T., P. Lougovski, K.S., E.D., and I.H. acknowledge funding from the U.S. Department of Energy (DOE), Office of Science, Office of Advanced Scientific Computing Research (ASCR) Quantum Computing Application Teams program, under fieldwork Proposal No. ERKJ347 and the Accelerated Research in Quantum Computing program under Awards No. DE-SC0020316 and No. DE-SC0025509. P. Lotshaw was supported at ORNL by the Defense Advanced Research Project Agency, Defense Science Office under Contract No. HR001120C0046 with Georgia Tech Research Institute.

This work has been partially supported by U.S. DOE Grant No. DE-FG02-13ER41967. ORNL is managed by

UT-Battelle, LLC, under Contract No. DE-AC0500OR22725 for the U.S. Department of Energy. The U.S. Government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others to do so, for U.S. Government purposes.

DATA AVAILABILITY

The data that support the findings of this article are not publicly available. The data are available from the authors upon reasonable request. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan [76].

APPENDIX A: NONUNIFORM PRECISION ERRORS

Here, we extend the model presented in Sec. III B to nonuniform precision errors. Consider a generic mixer Hamiltonian $H_M = \sum_i H_{M,i}$ and cost Hamiltonian $H_C = \sum_i H_{C,i}$. Nonuniform precision errors would affect each term distinctly such that

$$H_E(t) = [1 - s(t)] \sum_i \eta_{M,i}(t) H_{M,i} + s(t) \sum_i \eta_{C,i}(t) H_{C,i}. \tag{A1}$$

For example, if we consider the typical form of H_C as an Ising model, the nonuniform case would allow for distinct errors on local fields and coupling strengths.

Each error function $\eta_{M,i}(t)$ and $\eta_{C,i}(t)$ can be associated with its own statistical properties. Thus, the mean is given by $\overline{\eta_{\mu,i}(t)} = \eta_{\mu,i}$, $\mu = M, C$. Similarly, the two-point correlation, assuming no cross-correlations, can be described by $\overline{\eta_{\mu,i}(t_1)\eta_{\nu,j}(t_2)} = \delta_{\mu\nu}\delta_{ij}\Gamma_{\mu,i}(t_1 - t_2)$. We will now show that the effect of this alteration can be accounted for in the existing cumulant expressions in Sec. IV in a straightforward manner.

First, consider the case of coherent errors, as discussed in Sec. IV C. The generalization still focuses on the first cumulant; however, the rotated-frame Hamiltonian now takes the form

$$\tilde{H}_\mu(t) = \sum_i \eta_{\mu,i}(t) H_\mu(t). \tag{A2}$$

The error dependence is moved within the sum, with $H_\mu(t) = U_0(T,t) H_\mu U_0^\dagger(T,t)$. Utilizing this definition, we find the updated first-cumulant expression:

$$\mathcal{C}_1(T) = \sum_{j=1}^{p} \left(\beta_j \sum_k \eta_{M,k} \mathcal{H}^{(1,k)}_{M,j+1} + \gamma_j \sum_k \eta_{C,k} \mathcal{H}^{(1,k)}_{C,j} \right). \tag{A3}$$

The conjugated Hamiltonian is now given by $\mathcal{H}^{(1,k)}_{\mu,j} = \mathcal{Q}_j(H_{\mu,k}) - O^{-1}\mathcal{Q}_j(H_{\mu,k})O$.

Following a similar procedure, we calculate the error contribution for stochastic precision errors. Here, the first cumulant is zero and the second cumulant dominates. By introducing the updated definition of the cost and mixer Hamiltonians into the second cumulant, we obtain a modified version of Eq. (26):

$$\mathcal{C}_2(T) = \sum_{j=1}^{p} \beta_j \sum_k \frac{\Gamma_{M,k}}{2} \mathcal{H}^{(2,k)}_{M,j+1} + \sum_{j=1}^{p} \sum_k \frac{\Gamma_{C,k}}{2} \gamma_j \mathcal{H}^{(2,k)}_{C,j}, \tag{A4}$$

where $\mathcal{H}^{(2,k)}_{\mu,j} = \mathcal{Q}_j(H^2_{\mu,k}) + O^{-1}\mathcal{Q}_j(H^2_{\mu,k})O - 2O^{-1}\mathcal{Q}_j(H_{\mu,k}) O\mathcal{Q}_j(H_{\mu,k})$.

Depending upon the relationship between the statistical properties, it is possible for the uniform case to lower-bound the nonuniform case. Trivially, if $\eta_\mu = \min_i \eta_{\mu,i}$, then the norm of the first cumulant in the nonuniform case will be lower bounded by the norm of the first cumulant in the uniform case. More generally, if $\eta_{\mu,i} \approx \epsilon \ll 1$, which would be consistent with weak noise, then any $\eta_\mu < K\epsilon$ would suffice, where K is the number of terms in H_M or H_C. A similar argument can be made for Γ_μ, $\Gamma_{\mu,i}$ as well. Regardless of the conditions on the noise parameters, the scaling remains the same. Coherent errors result in unwanted unitary evolution, while stochastic errors yield exponential reductions in algorithmic performance.

APPENDIX B: ALTERNATIVE TOGGLING FRAME

Faulty QAOA evolution dictated by Eq. (17) can be factorized according to the rotating frame discussed in Sec. IV A or, equivalently, $U(T) = U_0(T)\tilde{U}'_E(T)$. The latter expression follows a very typical interaction picture representation in which the time-evolution operator,

$$\tilde{U}'_E(T) = \mathcal{T}_+ e^{-i\int_0^T \tilde{H}'_E(t)\,dt}, \tag{B1}$$

is governed by $\tilde{H}'_E(t) = U_0^\dagger(t) H_E(t) U_0(t)$. Equivalence between the two representations can be shown in a very straightforward manner by observing that

$$\begin{aligned} U(T) &= U_0(T)\tilde{U}'_E(T) \\ &= U_0(T)\tilde{U}'_E(T)U_0^\dagger(T)U_0(T) \\ &= \tilde{U}_E(T)U_0(T), \end{aligned} \tag{B2}$$

thus matching the rotating frame introduced in Sec. IV A.

Both representations lend themselves to cumulant-based expansions of expectation values with the latter inducing a time dependence on the observable. Following the approach utilized in the main text, the expectation value of an observable O can be expressed in terms of $U(T) = U_0(T)\tilde{U}'_E(T)$ as

$$\begin{aligned} \langle O \rangle &= \overline{\mathrm{Tr}[O\rho(T)]} \\ &= \overline{\mathrm{Tr}[OU_0(t)U_E(t)\rho(0)(\tilde{U}'_E(t))^\dagger U_0^\dagger(t)]} \\ &= \mathrm{Tr}[\Lambda(t)\rho(0)\tilde{O}(t)]. \end{aligned} \tag{B3}$$

The error dynamics are determined by the operator $\Lambda(t) = \overline{\tilde{O}^{-1}(t)U_E^\dagger(t)\tilde{O}(t)U_E(t)}$, which includes a conjugation by $\tilde{O}(t) = U_0^\dagger(t) O U_0(t)$, the observable in the interaction picture with respect to the noiseless QAOA evolution. As can be seen from comparing the above expressions to those surrounding Eq. (21), the distinction lies in the definition of the observable. Note that while both representations are analogous, the additional time dependence can result in unnecessary complexity in analytical or numerical investigations. Hence, we select the former for this study.

APPENDIX C: CUMULANT EXPANSION OF THE ERROR OPERATOR

Formally, the error operator can be expressed in terms of a cumulant expansion. This is observed by rewriting the error operator as [38]

$$\Lambda(T) = \overline{\mathcal{T}_+ e^{-i\int_{-T}^{T}\tilde{H}_{E,O}(t)dt}}, \tag{C1}$$

where the observable-dependent effective Hamiltonian is given by

$$\tilde{H}_{E,O}(t) = \begin{cases} \tilde{H}_E(T+t), & t\in[-T,0] \\ -O^{-1}\tilde{H}_E(T-t)O, & t\in[0,T]. \end{cases} \tag{C2}$$

Equation (C1) lends itself to the cumulant expansion in which the cumulant expressions can be obtained from the moment-generating equation

$$\Lambda(T) = \overline{\mathcal{T}_+ e^{-i\int_{-T}^{T}\tilde{H}_{E,O}(t)dt}} = e^{\mathcal{C}(T)}. \tag{C3}$$

In general, the expansion of $\Lambda(T)$ involves an infinite number of terms. While in certain scenarios the above cumulant expansion can be truncated *exactly* [38], this will not be possible for the faulty QAOA problem. As a result, we will truncate the expansion to capture the *approximate* error dynamics. Leveraging the weak-noise condition discussed above, the error dynamics are assumed to be well approximated by a second-order truncation on $\Lambda(T)$. More concretely, it is assumed that

$$\|\mathcal{C}(T)\|_\infty \simeq \left\|\mathcal{C}_1(T) + i\tfrac{1}{2}\mathcal{C}_2(T)\right\|_\infty. \tag{C4}$$

Note that, through the triangle inequality, it is trivial to show that the truncated norm of the cumulant expansion can be upper bounded as $\|\mathcal{C}(T)\|_\infty \lesssim \|H_E(T)\|_\infty + \|H_E(T)\|_\infty^2$, thus explicitly conveying a connection between the cumulant expansion and the weak-noise condition.

Under the above condition, the error dynamics are characterized by the first and second cumulants. The first-order term

$$\mathcal{C}_1(T) = \int_0^T dt_1[C^{(1)}(\tilde{H}_E(t_1)) - C^{(1)}(\tilde{H}'_E(t_1))] \tag{C5}$$

is dependent upon the rotated-frame error Hamiltonian, its observable-conjugated counterpart $\tilde{H}'_E(t) = O^{-1}\tilde{H}_E(t)O$, and the cumulant expression $C^{(1)}(A) = \overline{A}$. The second-order term,

$$\frac{\mathcal{C}_2(T)}{2!} = I_1(T) + I_2(T) - I_3(T) - I_4(T), \tag{C6}$$

can be decomposed into four integrals:

$$I_1(T) = \int_0^T dt_1 \int_0^{t_1} dt_2\ C^{(2)}(\tilde{H}_E(t_1), \tilde{H}_E(t_2)), \tag{C7}$$

$$I_2(T) = \int_0^T dt_1 \int_0^{t_1} dt_2\ C^{(2)}(\tilde{H}'_E(t_2), \tilde{H}'_E(t_1)), \tag{C8}$$

$$I_3(T) = \int_0^T dt_1 \int_0^{t_1} dt_2\ C^{(2)}(\tilde{H}'_E(t_1), \tilde{H}_E(t_2)), \tag{C9}$$

$$I_4(T) = \int_0^T dt_1 \int_0^{t_1} dt_2\ C^{(2)}(\tilde{H}'_E(t_2), \tilde{H}_E(t_1)). \tag{C10}$$

Each integral includes the second-order cumulant expression

$$C^{(2)}(A,B) = \overline{AB} - \tfrac{1}{2}(\overline{A}\,\overline{B} + \overline{B}\,\overline{A}). \tag{C11}$$

Together, these equations form the quantum control framework that is exploited to calculate the approximate error dynamics generated by $H_E(t)$ and, subsequently, the expectation value of H_C in the presence of precision errors.

APPENDIX D: ADDITIONAL BOUNDS

1. Absolute error bound: Noninvertible observable

Consider the case where an observable O is noninvertible; however, it can be expressed as a sum of invertible operators O_i: $O = \sum_i O_i$. A trivial example of an operator expansion of this type is the n-qubit Pauli basis for an operator $O \in SU(n)$. Of course, this particular example is relevant to the Grover problem and Ising-type problem considered in the main text and, more generally, in the context of variational quantum algorithms.

Utilizing this expansion for O, the expectation value of the observable can be written as

$$\begin{aligned}\overline{\langle O(T)\rangle} &= \sum_i \overline{\mathrm{Tr}[\rho(T)O_i]} \\ &= \sum_i \mathrm{Tr}[\Lambda_i(T)\rho_S(T)O_i],\end{aligned} \tag{D1}$$

where $\Lambda_i(T) = \overline{O_i^{-1}\tilde{U}_E^\dagger(T)O_i\tilde{U}_E(T)}$. Going a bit further and leveraging the cumulant expansion,

$$\overline{\langle O(T)\rangle} = \sum_i \mathrm{Tr}[e^{\mathcal{C}_{O_i}(T)}\rho_S(T)O_i], \tag{D2}$$

with $\mathcal{C}_{O_i}(T)$ representing the cumulant expansion containing the jth observable.

The above expressions can be employed to derive bounds on the various quantities examined in Sec. V of the main text. For example, consider the absolute error in the expectation value $|\overline{\Delta\langle O(T)\rangle}|$. An upper bound on this quantity can be derived following a similar approach to that of Sec. V A to yield

$$\begin{aligned}|\overline{\Delta\langle O(T)\rangle}| &= \left|\mathrm{Tr}\left[\left(\sum_i O_i\Lambda_i(T) - O\right)\rho_S(T)\right]\right| \\ &\overset{(1)}{\leqslant} \Big\|\sum_i O_i\Lambda_i(T) - O\Big\|_\infty\ \|\rho_S(T)\|_1 \\ &\overset{(2)}{\leqslant} \sum_i \|O_i\|_\infty \|\Lambda_i(T) - I\|_\infty \\ &\overset{(3)}{\leqslant} \sum_i \|O_i\|_\infty (e^{\|\mathcal{C}_{O_i}(T)\|_\infty} - 1).\end{aligned} \tag{D3}$$

Here, the variant of Hölder's inequality from Sec. V A is used to obtain (1). The trace norm of a density matrix is unity and, thus, the resulting bound in (2) is solely dependent upon the infinity norm of a term proportional to the accumulated error resulting from $\Lambda_i(T)$. In addition to the property of the density matrix, we apply the triangle inequality and submultiplicativity to obtain (2). By expressing the error operator as a cumulant expansion and bounding the error term as in Sec. E 1, we obtain the final bound in (3).

2. Bounds on cumulants

Many of the bounds derived above share a common feature: dependence on the operator norm of the error operator. Through Eq. (28), we find that the error dynamics can be bounded in terms of the operator norm of the cumulant series $\mathcal{C}(T)$. When the noise is sufficiently weak, the cumulant series is well approximated by the leading-order cumulant expression, i.e., $\mathcal{C}_1(T)$ or $\mathcal{C}_2(T)$ in the case of constant coherent errors or zero-mean, Gaussian stochastic errors, respectively. Here, we provide bounds on the first and second cumulants, which can be used to further upper-bound the expressions presented above.

Using the expressions derived in Sec. IV for the first and second cumulants, bounds are obtained for each noise scenario. In the case of constant coherent errors, the first cumulant [Eq. (C5)] can be shown to be bounded according to

$$\|\mathcal{C}_1\| \leqslant \sum_{\mu=M,C} \eta_\mu \sum_{j=1}^{p} \left|g_j^\mu\right| \|H_\mu\| (1 + \|O^{-1}\| \|O\|). \tag{D4}$$

Utilizing similar techniques, such as the triangle inequality, submultiplicativity, and unitary invariance, a similar expression can be obtained for the second cumulant [Eq. (C6)]:

$$\begin{aligned}\|\mathcal{C}_2\| &\leqslant \|I_1(T)\| + \|I_2(T)\| + 2\|I_3(T)\| \\ &\leqslant \sum_{\mu=M,C} \Gamma_\mu \sum_{j=1}^{p} \left|g_j^\mu\right| \|H_\mu\| (1 + 3\|O^{-1}\| \|O\|).\end{aligned} \tag{D5}$$

For brevity and without loss of generality, the subscript denoting the operator norm has been dropped and bounds are displayed in terms of a generic unitary invariant norm.

Intuitively, bounds on the first and second cumulants scale with mean and variance, respectively. Both bounds are further characterized by the accumulative time over which each QAOA constituent Hamiltonian contributes to the evolution. Additional dependence on the norm of the constituent Hamiltonians and the observable O and its inverse are also observed, the latter emerging via integral expressions conjugated by O.

3. Bounds on gradients of cumulants

Here, we include bounding expressions for $\|\partial_{\gamma_k}\mathcal{C}_n(T)\|$ given their relevance to the gradient bound derived in Eq. (38). While we focus of the derivative with respect to γ_k, we note that equivalent expressions for the derivative with respect to β_k follow directly. The primary difference is a change of QAOA Hamiltonian from H_C to H_M at the appropriate locations.

In order to bound the derivatives of the cumulants, we begin by noting that

$$\partial_{\gamma_k} Q_{p:j+1} = \begin{cases} -iQ_{p:k}H_C Q_{k-1:j+1}, & k \geqslant j+1 \\ 0, & \text{otherwise.} \end{cases} \tag{D6}$$

This expression can be utilized to bound on the derivative of the rotated-frame Hamiltonians as

$$\|\partial_{\gamma_k}(Q_{p:j+1}H_C Q_{p:j+1}^\dagger)\| \leqslant 2\|H_C\|^2, \tag{D7}$$

for $k \geqslant j+1$, which in turn can be employed to bound both cumulants. The first cumulant is bounded according to

$$\begin{aligned}\left\|\partial_{\gamma_k}\mathcal{C}_1(T)\right\| &\leqslant (1 + \|O^{-1}\|\|O\|)\|H_C\| \\ &\times \left[2\eta_C\|H_M\| \sum_{j=1}^{k-1}|\beta_j| \right. \\ &\left. + \eta_M\left(1 + 2\|H_C\| \sum_{j=1}^{k}|\gamma_j|\right)\right],\end{aligned} \tag{D8}$$

where the sums over $j \leqslant k-1$ arise from Eq. (D6). The derivative of the second cumulant is bounded as

$$\begin{aligned}\left\|\partial_{\gamma_k}\mathcal{C}_2(T)\right\| &\leqslant \left\|\partial_{\gamma_k}I_1(T)\right\| + \left\|\partial_{\gamma_k}I_2(T)\right\| + 2\left\|\partial_{\gamma_k}I_3(T)\right\| \\ &\leqslant \left\|\partial_{\gamma_k}I_1(T)\right\|(1 + \|O^{-1}\|\|O\|) \\ &+ 2\left\|\partial_{\gamma_k}I_3(T)\right\|,\end{aligned} \tag{D9}$$

in terms of its components integrals. In turn, these integrals can be further bounded in terms of the variational parameters and the norm of the mixer and problem Hamiltonian. The first integral bound is

$$\begin{aligned}\left\|\partial_{\gamma_k}I_1(T)\right\| &\leqslant \Gamma_M\|H_C\|^2\left(1 + 2\|H_C\|\sum_{j=1}^{k}|\gamma_j|\right) \\ &+ 2\Gamma_C\|H_M\|^2\|H_C\|\sum_{j=1}^{k-1}|\beta_j|,\end{aligned} \tag{D10}$$

while the derivative of the third integral is bounded by

$$\begin{aligned}\left\|\partial_{\gamma_k}I_3(T)\right\| &\leqslant \|O^{-1}\|\|O\|\|H_C\| \\ &\times \left[\Gamma_C\|H_C\|\left(1 + 4\|H_C\|\sum_{j=1}^{k}|\gamma_j|\right)\right. \\ &\left. + 4\Gamma_M\|H_M\|^2\sum_{j=1}^{k-1}|\beta_j|\right].\end{aligned} \tag{D11}$$

4. Finite sampling effects

Finite sampling of expectation values will result in estimates that deviate from the asymptotic, infinite sample mean. As such, we account for finite sampling effects in the bounds derived in Sec. V by modifying the estimator in accordance with the central limit theorem to

$$\langle O\rangle_{\text{est}} = \langle O\rangle \pm \frac{1}{\sqrt{N_s}}\sqrt{\text{Var}(O)}. \tag{D12}$$

By definition, this assumes that the sampling statistics of the expectation value are Gaussian. We make use of Eq. (D12) below to derive finite sampling corrections to the bounds on absolute error and mean-square error.

a. Absolute error

Here, we attain a bound on the absolute error between the finitely sampled noisy expectation value $\langle O\rangle_{\text{est}}$ and the asymptotic noiseless expectation value $\langle O\rangle_0$. Following Eq. (27), this quantity is defined as

$$|\overline{\Delta\,\langle O(T)\rangle}_{\text{est}}| = |\overline{\langle O(T)\rangle}_{\text{est}} - \langle O(T)\rangle_0\,|. \tag{D13}$$

Using Eq. (D12), this expression can be bounded by the sum of the asymptotic bound for $|\overline{\Delta\,\langle O(T)\rangle}|$ [Eq. (28)] and a correction term proportional to $1/\sqrt{N_s}$. More concretely, it can be shown that

$$\begin{aligned}|\overline{\Delta\,\langle O(T)\rangle}_{\text{est}}| - |\overline{\Delta\,\langle O(T)\rangle}| &\leqslant \frac{1}{\sqrt{N_s}}|\sqrt{\text{Var}(O)}| \\ &\leqslant \frac{1}{\sqrt{N_s}}\sqrt{\kappa(O,T)},\end{aligned} \tag{D14}$$

where

$$\kappa(O,T) = \|\Lambda(T)\|_\infty\|O^2\rho_0(T)\|_1 + \|\Lambda(T)\|_\infty^2\|O\rho_0(T)\|_1^2 \tag{D15}$$

results from upper-bounding the variance of O.

b. MSE

Following Eq. (29), the MSE in the finite sampling regime is defined as

$$\overline{\text{MSE}(O(T))}_{\text{est}} = \overline{\text{Var}_{\text{est}}(O(T))} + (\overline{\Delta\,\langle O(T)\rangle}_{\text{est}})^2, \tag{D16}$$

where the estimator is now designated as the finitely sampled expectation value given in Eq. (D12). Similar to the bound on the bias developed in Eq. (D14), the variance can be shown to be bounded by the sum of the asymptotic variance and N_s-dependent corrections, i.e.,

$$\begin{aligned}\text{Var}_{\text{est}}(O) &= \langle O^2\rangle_{\text{est}} - (\langle O\rangle_{\text{est}})^2 \\ &\leqslant \text{Var}(O) + \frac{1}{N_s}\text{Var}(O) \\ &\quad + \frac{1}{\sqrt{N_s}}[\sqrt{\text{Var}(O^2)} + \langle O\rangle\sqrt{\text{Var}(O)}].\end{aligned} \tag{D17}$$

The relative difference between the finitely sampled and asymptotic variance can be further bounded according to

$$\begin{aligned}&\overline{\text{Var}_{\text{est}}(O)} - \overline{\text{Var}(O)} \\ &\leqslant \frac{1}{N_s}\kappa(O,T) + \frac{1}{\sqrt{N_s}}[\kappa(O^2,T) \\ &\quad + 2\|\Lambda(T)\|_\infty\|O\rho_0(T)\|_1|\sqrt{\kappa(O,T)}].\end{aligned} \tag{D18}$$

As expected by the definition of Eq. (D12), the dominant term scales according to $1/\sqrt{N_s}$.

APPENDIX E: DERIVATIONS

Here, we provide additional details regarding the bounds presented in Sec. V.

1. Bound on error operator

In Sec. V A, the operator norm of the operator $\Lambda(T)$ was said to be bounded by the exponentiated norm of the cumulant series. We prove this bound as follows:

$$\begin{aligned}\|\Lambda(T)\| &= \|e^{\mathcal{C}(T)}\| \\ &= \left\|\sum_{m=0}^{\infty}\mathcal{C}^m(T)/m!\right\| \\ &\overset{(1)}{\leqslant} \sum_{m=0}^{\infty}\|\mathcal{C}^m(T)/m!\| \\ &\overset{(2)}{\leqslant} \sum_{m=0}^{\infty}\|\mathcal{C}(T)\|^m/m! \\ &= e^{\|\mathcal{C}(T)\|}.\end{aligned} \tag{E1}$$

Here, (1) the triangle inequality and (2) submultiplicativity have been used to obtain the upper-bounding expressions.

2. Bound on MSE: Coherent error

MSE can be expressed as a sum of the variance and the bias of the estimator. Here, we derive bounds on each term individually for the case of constant coherent errors. The resulting bounds are combined to obtain Eq. (30). Although the bound derived in the subsequent section encompasses constant coherent errors, we take an alternative approach that leverages results from Trotter error analysis to obtain a tighter bound for a case where ensemble averaging (i.e., a cumulant expansion) is not required.

A bound on the variance can be obtained by first noting that

$$\begin{aligned}\text{Var}(O(T)) &= \text{Tr}[\rho(T)O^2] - \text{Tr}[\rho(T)O]^2 \\ &\leqslant |\text{Tr}[\rho(T)O^2]| + |\text{Tr}[\rho(T)O]|^2,\end{aligned} \tag{E2}$$

where we have used $\text{Var}(O) = |\text{Var}(O)|$ and the triangle inequality. Employing Hölder's inequality and submultiplicativity, this bound is further reduced to

$$\begin{aligned}\text{Var}(O(T)) &\leqslant \|O^2\|_\infty + \|O^2\|_\infty^2 \\ &\leqslant 2\|O\|_\infty^2.\end{aligned} \tag{E3}$$

Note that the first inequality makes use of $\|\rho(T)\|_1 = 1$.

The squared bias is bounded by first bounding the difference between the faulty and noiseless expectation values. Following a similar procedure to that utilized above,

$$\begin{aligned}|\Delta\,\langle O(T)\rangle\,| &= |\text{Tr}[\rho(T) - \rho_0(T)O]| \\ &= |\text{Tr}[(U^\dagger(T)OU(T) - U_0^\dagger(T)OU_0(T))\rho(0)]| \\ &\overset{(1)}{\leqslant} \|(U^\dagger(T)OU(T) - U_0^\dagger(T)OU_0(T))\|_\infty.\end{aligned} \tag{E4}$$

We now introduce an additional term, $U^\dagger(T)OU_0(T) - U^\dagger(T)OU_0(T)$, within the operator norm and group terms associated with the difference $U^\dagger(T) - U_0(T)$ and its Hermitian conjugate. Using unitary invariance of the norm and submultiplicativity, we obtain

$$|\Delta\,\langle O(T)\rangle\,| \leqslant 2\|O\|_\infty\|U(T) - U_0(T)\|_\infty. \tag{E5}$$

The norm of the difference between the faulty and noiseless propagators is further bounded through the definition of the

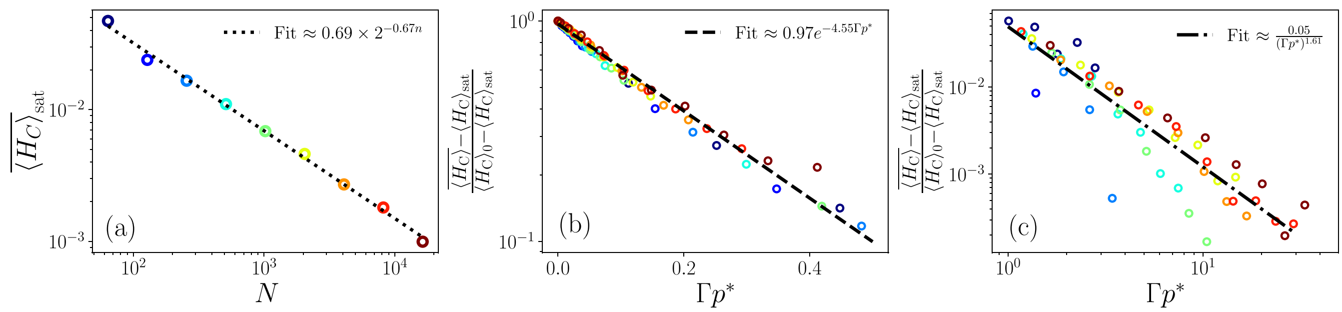

FIG. 8. Scaling fits for the expectation value of the Grover cost Hamiltonian in the presence of stochastic errors [data corresponding to Fig. 4(a)]. (a) Fit to the saturation value for large noise. The saturation value decreases linearly with problem size, or exponentially with the number of qubits. (b) Weak-noise ($\Gamma p^* < 0.5$) behavior of the decay function [see Eq. (F2)]. For weak noise, the expectation value decays exponentially, as evident from the linear fit on the log-linear plot. (c) Large-noise ($\Gamma p^* > 1$) behavior of the decay function which behaves as a power law.

time-evolution dynamical equation. Namely,

$$\begin{aligned}\|U(T) - U_0(T)\|_\infty &= \|\tilde{U}_E(T) - 1\|_\infty \\ &= \left\| -i \int_0^T \tilde{H}_E(t) U_E(t)\, dt \right\|_\infty \\ &\leqslant \int_0^T \|H_E(t)\|_\infty\, dt \\ &\leqslant h_E, \end{aligned} \tag{E6}$$

where the triangle inequality and submultiplicativity are used to achieve the final bound in terms of Eq. (31). Including Eq. (E6) in Eq. (E5) leads to $(\Delta\langle O(T)\rangle)^2 \leqslant 4\|O\|_\infty^2 h_E^2$, which in turn yields Eq. (30) when combined with Eq. (E3).

3. Bound on exact gradient

In Sec. V C, we briefly derived a bound on the absolute error in the gradient of the expectation value for a single variational parameter. We elaborate on that derivation here, focusing specifically on the expressions related to norms comprised of the derivative of the error operator and the derivative of the ideal QAOA time-evolved state. The former appears in the bound derived in Eq. (38), while the latter is a part of Eq. (39).

A bound on the norm of the derivative of the error operator is shown in Eq. (38). We elucidate the details on the derivation of this bound as follows:

$$\begin{aligned}\|\partial_{\gamma_k}\Lambda(T)\| &= \left\| e^{\mathcal{C}(T)} \int_0^1 d\lambda\; e^{-\lambda\mathcal{C}(T)} \partial_{\gamma_k}\mathcal{C}(T) e^{\lambda\mathcal{C}(T)} \right\| \\ &\leqslant \|e^{\mathcal{C}(T)}\| \int_0^1 d\lambda \left\| e^{-\lambda\mathcal{C}(T)} \partial_{\gamma_k}\mathcal{C}(T) e^{\lambda\mathcal{C}(T)} \right\| \\ &\leqslant e^{\|\mathcal{C}(T)\|} \int_0^1 e^{2\lambda\|\mathcal{C}(T)\|} d\lambda\; \left\|\partial_{\gamma_j}\mathcal{C}(T)\right\| \\ &= \frac{e^{\|\mathcal{C}(T)\|}}{2\|\mathcal{C}(T)\|}(e^{2\|\mathcal{C}(T)\|} - 1)\left\|\partial_{\gamma_k}\mathcal{C}(T)\right\|. \end{aligned} \tag{E7}$$

The first equality is obtained by expressing the error operator as a cumulant expansion, i.e., $\Lambda(T) = e^{\mathcal{C}(T)}$, and then using an identity derived by Snider [77] to formally calculate the derivative of an exponential operator. Subsequent inequalities follow from applications of submultiplicativity and Eq. (E1), with the final equality resulting from an evaluation of the integral.

While the first term in the bound given in Eq. (36) is proportional to the derivative of the error operator, the second is proportional to $\partial_{\gamma_k}\rho_0(T)$. The subsequent bound on this term [Eq. (39)] is first computed by noting that

$$\begin{aligned}\partial_{\gamma_k}\rho_0(T) &= (\partial_{\gamma_k}U)\rho(0)U^\dagger + U\rho(0)(\partial_{\gamma_k}U^\dagger) \\ &= -iQ_{p:k+1}[H_C, Q_{k:1}\rho(0)Q_{k:1}^\dagger]Q_{p:k+1}^\dagger \\ &= -iQ_{p:k+1}[H_C, \rho(T_k)]Q_{p:k+1}^\dagger. \end{aligned} \tag{E8}$$

Using Hölder's inequality to bound the second term of Eq. (36), we obtain an expression that contains a norm of the product of O and $\partial_{\gamma_k}\rho_0(T)$. Employing the result from Eq. (E8), one can show

$$\begin{aligned}\|O\partial_{\gamma_k}\rho_0(T)\| &= \|OQ_{p:k+1}[H_C, \rho(T_k)]Q_{p:k+1}^\dagger\| \\ &= \|Q_{p:k+1}^\dagger OQ_{p:k+1}[H_C, \rho(T_k)]\| \\ &= \|\tilde{O}(T, T_k)[H_C, \rho(T_k)]\|. \end{aligned} \tag{E9}$$

Note that the second equality follows from the unitary invariance of the norm.

APPENDIX F: ADDITIONAL RESULTS FOR NUMERICAL SIMULATIONS

In this section, we provide additional details regarding the numerical simulations discussed in Sec. VI. In Appendix F 1 we discuss additional results for the Grover problem and in Appendix F 2 we provide additional information on the Ising problem.

1. Grover problem

In this section we provide additional details for the fitting procedure used to extract the exponents discussed in Sec. VI A. We discuss the behavior of the noise-averaged expectation value in the presence of stochastic errors and coherent errors in Figs. 8 and 9, respectively. Finally, in Fig. 10 we discuss the fits to the distance between noisy and noiseless QAOA evolution.

Let us start by discussing the case of stochastic precision errors for the optimal Grover QAOA, the data for which are

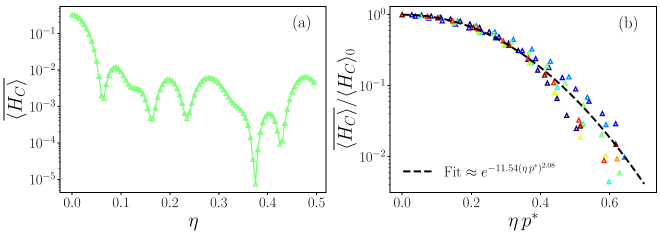

FIG. 9. Scaling fits for the expectation value of the Grover cost function in the presence of coherent error [data corresponding to Fig. 4(b)]. (a) The nonmonotonic behavior of the noisy expectation value for increasing magnitude of the coherent error. (b) The decay of the expectation value for small coherent error $\eta p^* \lesssim 1$ behaves approximately as a squared exponential decay.

in Fig. 4(a) in the main text. The results of the fitting procedure are discussed in Fig. 8. We fit the numerically evaluated noisy expectation value to the following phenomenological functional form that models both weak- and strong-noise behavior:

$$\overline{\langle H_C\rangle} = \langle H_C\rangle_0\, \mathcal{S}(p^*) + \langle H_C\rangle_{\text{sat}}\, [1 - \mathcal{S}(p^*)]. \tag{F1}$$

This functional form allows us to extract the decay function from the expression

$$\mathcal{S}(p^*) = \frac{\overline{\langle H_C\rangle} - \langle H_C\rangle_{\text{sat}}}{\langle H_C\rangle_0 - \langle H_C\rangle_{\text{sat}}}, \tag{F2}$$

with the condition $\mathcal{S}(p^* = 0) = 1$. In Fig 8(a), we extract the functional form of the saturation value by examining the large-noise behavior ($\Gamma = 1$). We find that the saturation value decays exponentially with the number of qubits, n (or linearly with problem size N):

$$\overline{\langle H_C\rangle}_{\text{sat}} \approx 0.69 \times 2^{-0.67n}. \tag{F3}$$

Utilizing this expression for the large-noise saturation value, we obtain a fit for the decay function, $\mathcal{S}(\Gamma, p^*)$, in Figs. 8(b) and 8(c). We find two distinct functional behaviors for the weak noise [Fig. 8(b)] and large noise [Fig. 8(c)], respectively:

$$\mathcal{S}(p^*) = \begin{cases} 0.97e^{-4.55\Gamma p^*}, & \Gamma p^* < 0.5 \text{ [weak noise]} \\ \frac{0.05}{(\Gamma p^*)^{1.61}}, & \Gamma p^* > 1.0 \text{ [strong noise]}. \end{cases} \tag{F4}$$

Note that the weak- and strong-noise behavior is closely related to the behavior of the scaling variable Γp^*.

Next, we extract the scaling behavior of the noisy expectation value in the presence of coherent errors, the data for

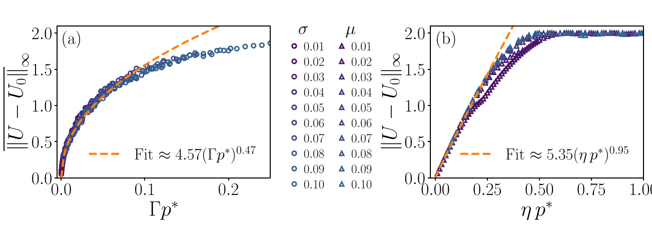

FIG. 10. Scaling fits for the norm difference with increasing number of layers and noise, data corresponding to Figs. 3(c) and 3(d). (a) Diffusive growth in the distance for stochastic precision error. (b) Linear growth in the distance for coherent precision error.

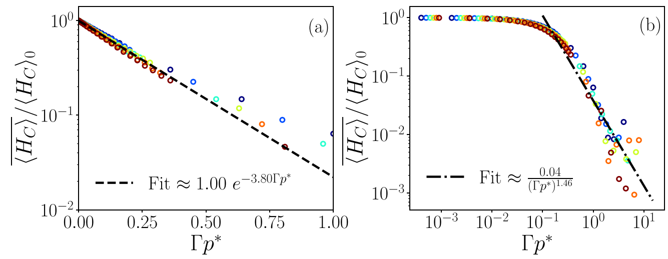

FIG. 11. Scaling fits for GHZ Hamiltonian expectation value for stochastic errors. (a) For weak noise ($\Gamma p^* \lesssim 1$), the expectation value of the Hamiltonian decays exponentially. (b) For strong noise ($\Gamma p^* \gtrsim 1$), the expectation value decays as a power law.

which are shown in Fig. 4(b) in the main text. The scaling fit is discussed in Fig. 9. We note that the measured expectation value in the presence of constant coherent error does not show a monotonic behavior with increase in errors. This is shown in Fig. 9(a), where we see that, after an initial decay in the expectation value, the expectation value exhibits oscillations. We extract the initial exponential decay by examining the cases where $\eta p^* \lesssim 1$. The initial decay is well described by an exponential decay:

$$\overline{\langle H_C\rangle} = \langle H_C\rangle_0\, S(p^*), \tag{F5}$$

$$\mathcal{S}(p^*) = e^{-11.54(\eta\, p^*)^{2.08}}. \tag{F6}$$

Note that the expectation value decays as a squared exponential.

Finally, let us discuss the scaling of the difference between the noisy and noiseless QAOA evolution operator. We numerically evaluate the ∞-norm of the average difference between the noisy unitary U and the noiseless one, U_0, in Fig. 10, the data for which correspond to Figs. 3(c) and 3(d). We have the following behavior for weak noise or small p ($p\Gamma \ll 1, \eta p \ll 1$):

$$\overline{\|U - U_0\|_\infty} \approx \begin{cases} 4.57(\Gamma p)^{0.47}, & \text{[stochastic error]} \\ 5.35(\eta p)^{0.95}, & \text{[coherent error]}. \end{cases} \tag{F7}$$

This scaling is consistent with the norm difference increasing in a diffusive manner for stochastic error ($\propto \sqrt{\Gamma p^*}$) and linearly for coherent error ($\propto \eta p^*$).

2. Ising problem

In this section, we elaborate on the scaling fits for the Ising problem on a ring, which is discussed in Sec. VI B. The data correspond to Fig. 6 in the main text. We fit the decay in the expectation value to the following phenomenological function:

$$\langle H_C\rangle \approx \langle H_C\rangle_0\, \mathcal{S}(p^*). \tag{F8}$$

For stochastic error, as discussed in Fig. 11, the Hamiltonian expectation value decays as

$$\mathcal{S}(p^*) = \begin{cases} 1.00e^{-3.80\Gamma p^*}, & \Gamma p^* < 1.0 \text{ [weak noise]} \\ \frac{0.04}{(\Gamma p^*)^{1.46}}, & \Gamma p^* > 1.0 \text{ [strong noise]}. \end{cases} \tag{F9}$$

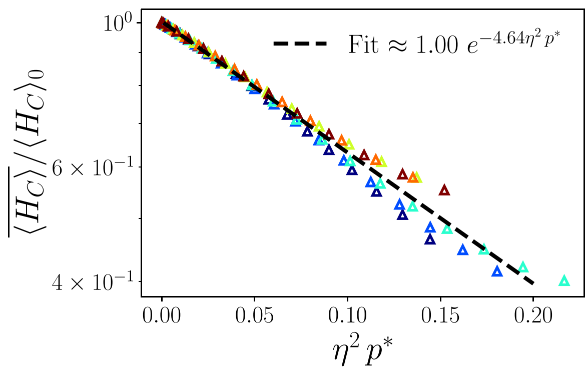

FIG. 12. Scaling fits for the expectation value of the cost function for GHZ state preparation in the presence of coherent error. The decay is consistent with an exponential decay.

Again, for stochastic errors, the expectation value decays exponentially for weak strength of the noise, then changes to a power law for larger strength of noise.

Next we discuss the effect of coherent errors. In this case, the decay function for weak noise is again consistent with an exponential decay,

$$\mathcal{S}(p^*) = e^{-4.64\eta^2 p^*}. \tag{F10}$$

Note that the exponential decay is consistent with a scaling variable $\eta^2 p^*$. The scaling fits for coherent errors are shown in Fig. 12.

[1] E. Farhi, J. Goldstone, and S. Gutmann, A quantum approximate optimization algorithm, arXiv:1411.4028.

[2] E. Farhi, J. Goldstone, and S. Gutmann, A quantum approximate optimization algorithm applied to a bounded occurrence constraint problem, arXiv:1412.6062.

[3] D. Wecker, M. B. Hastings, and M. Troyer, Training a quantum optimizer, Phys. Rev. A **94**, 022309 (2016).

[4] R. Biswas, Z. Jiang, K. Kechezhi, S. Knysh, S. Mandrà, B. O'Gorman, A. Perdomo-Ortiz, A. Petukhov, J. Realpe-Gómez, E. Rieffel, D. Venturelli, F. Vasko, and Z. Wang, A NASA perspective on quantum computing: Opportunities and challenges, Parallel Comput. **64**, 81 (2017).

[5] Z. Jiang, E. G. Rieffel, and Z. Wang, Near-optimal quantum circuit for Grover's unstructured search using a transverse field, Phys. Rev. A **95**, 062317 (2017).

[6] Z. Wang, S. Hadfield, Z. Jiang, and E. G. Rieffel, Quantum approximate optimization algorithm for MaxCut: A fermionic view, Phys. Rev. A **97**, 022304 (2018).

[7] S. Lloyd, Quantum approximate optimization is computationally universal, arXiv:1812.11075.

[8] E. Farhi and A. W. Harrow, Quantum supremacy through the quantum approximate optimization algorithm, arXiv:1602.07674.

[9] S. Hadfield, Z. Wang, B. O'Gorman, E. G. Rieffel, D. Venturelli, and R. Biswas, From the quantum approximate optimization algorithm to a quantum alternating operator ansatz, Algorithms **12**, 34 (2019).

[10] E. Farhi, J. Goldstone, S. Gutmann, and L. Zhou, The quantum approximate optimization algorithm and the Sherrington-Kirkpatrick model at infinite size, Quantum **6**, 759 (2022).

[11] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, and J. M. Gambetta, Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets, Nature (London) **549**, 242 (2017).

[12] G. Pagano, A. Bapat, P. Becker, K. S. Collins, A. De, P. W. Hess, H. B. Kaplan, A. Kyprianidis, W. L. Tan, C. Baldwin, L. T. Brady, A. Deshpande, F. Liu, S. Jordan, A. V. Gorshkov, and C. Monroe, Quantum approximate optimization of the long-range Ising model with a trapped-ion quantum simulator, Proc. Natl. Acad. Sci. U.S.A. **117**, 25396 (2020).

[13] A. Bengtsson, P. Vikstål, C. Warren, M. Svensson, X. Gu, A. F. Kockum, P. Krantz, C. Križan, D. Shiri, I.-M. Svensson, G. Tancredi, G. Johansson, P. Delsing, G. Ferrini, and J. Bylander, Improved success probability with greater circuit depth for the quantum approximate optimization algorithm, Phys. Rev. Appl. **14**, 034010 (2020).

[14] M. Harrigan *et al.*, Quantum approximate optimization of non-planar graph problems on a planar superconducting processor, Nat. Phys. **17**, 332 (2021).

[15] M. Alam, A. Ash-Saki, and S. Ghosh, Analysis of quantum approximate optimization algorithm under realistic noise in superconducting qubits, arXiv:1907.09631.

[16] J. Marshall, F. Wudarski, S. Hadfield, and T. Hogg, Characterizing local noise in QAOA circuits, IOP SciNotes **1**, 025208 (2020).

[17] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, Noise-induced barren plateaus in variational quantum algorithms, Nat. Commun. **12**, 1 (2021).

[18] C. Xue, Z.-Y. Chen, Y.-C. Wu, and G.-P. Guo, Effects of quantum noise on quantum approximate optimization algorithm, Chin. Phys. Lett. **38**, 030302 (2021).

[19] J. Kattemölle and G. Burkard, Ability of error correlations to improve the performance of variational quantum algorithms, Phys. Rev. A **107**, 042426 (2023).

[20] F. B. Maciejewski, F. Baccari, Z. Zimborás, and M. Oszmaniec, Modeling and mitigation of cross-talk effects in readout noise with applications to the Quantum Approximate Optimization Algorithm, Quantum **5**, 464 (2021).

[21] Y. Dong, X. Meng, L. Lin, R. Kosut, and K. B. Whaley, Robust control optimization for quantum approximate optimization algorithms, IFAC-PapersOnLine **53**, 242 (2020).

[22] R. Harris, M. W. Johnson, T. Lanting, A. J. Berkley, J. Johansson, P. Bunyk, E. Tolkacheva, E. Ladizinsky, N. Ladizinsky, T. Oh, F. Cioata, I. Perminov, P. Spear, C. Enderud, C. Rich, S. Uchaikin, M. C. Thom, E. M. Chapple, J. Wang, B. Wilson *et al.*, Experimental investigation of an eight-qubit unit cell in a superconducting optimization processor, Phys. Rev. B **82**, 024511 (2010).

[23] P. I. Bunyk, E. M. Hoskinson, M. W. Johnson, E. Tolkacheva, F. Altomare, A. J. Berkley, R. Harris, J. P. Hilton, T. Lanting, A. J.

Przybysz et al., Architectural considerations in the design of a superconducting quantum annealing processor, IEEE Trans. Appl. Supercond. **24**, 1 (2014).
[24] A. D. King, T. Lanting, and R. Harris, Performance of a quantum annealer on range-limited constraint satisfaction problems, arXiv:1502.02098.
[25] S. Boixo, T. Albash, F. M. Spedalieri, N. Chancellor, and D. A. Lidar, Experimental signature of programmable quantum annealing, Nat. Commun. **4**, 2067 (2013).
[26] A. Pearson, A. Mishra, I. Hen, and D. A. Lidar, Analog errors in quantum annealing: doom and hope, npj Quantum Inf. **5**, 107 (2019).
[27] T. Albash, V. Martin-Mayor, and I. Hen, Analog errors in Ising machines, Quantum Sci. Technol. **4**, 02LT03 (2019).
[28] A. D. King and C. C. McGeoch, Algorithm engineering for a quantum annealing platform, arXiv:1410.2628.
[29] E. Pelofske, G. Hahn, and H. Djidjev, Optimizing the spin reversal transform on the d-wave 2000q, in *2019 IEEE International Conference on Rebooting Computing (ICRC)* (IEEE, New York, 2019), pp. 1–8.
[30] P. Krantz, M. Kjaergaard, F. Yan, T. P. Orlando, S. Gustavsson, and W. D. Oliver, A quantum engineer's guide to superconducting qubits, Appl. Phys. Rev. **6**, 021318 (2019).
[31] J. C. Bardin, D. H. Slichter, and D. J. Reilly, Microwaves in quantum computing, IEEE J. Microwaves **1**, 403 (2021).
[32] S. Joshi and S. Moazeni, Scaling up superconducting quantum computers with cryogenic RF-photonics, J. Lightwave Technol. **42**, 166 (2023).
[33] Z.-C. Yang, A. Rahmani, A. Shabani, H. Neven, and C. Chamon, Optimizing variational quantum algorithms using Pontryagin's minimum principle, Phys. Rev. X **7**, 021027 (2017).
[34] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, and M. D. Lukin, Quantum approximate optimization algorithm: Performance, mechanism, and implementation on near-term devices, Phys. Rev. X **10**, 021067 (2020).
[35] J. Wurtz and P. J. Love, Counterdiabaticity and the quantum approximate optimization algorithm, Quantum **6**, 635 (2022).
[36] P. W. Shor, Fault-tolerant quantum computation, in *IEEE Proceedings of 37th Conference on Foundations of Computer Science* (Burlington, VT, USA, 1996), pp. 56–65.
[37] G. A. Paz-Silva and L. Viola, General transfer-function approach to noise filtering in open-loop quantum control, Phys. Rev. Lett. **113**, 250501 (2014).
[38] G. A. Paz-Silva, L. M. Norris, and L. Viola, Multiqubit spectroscopy of Gaussian quantum noise, Phys. Rev. A **95**, 022121 (2017).
[39] L. M. Norris, D. Lucarelli, V. M. Frey, S. Mavadia, M. J. Biercuk, and L. Viola, Optimally band-limited spectroscopy of control noise using a qubit sensor, Phys. Rev. A **98**, 032315 (2018).
[40] V. Frey, S. Mavadia, L. Norris, W. De Ferranti, D. Lucarelli, L. Viola, and M. Biercuk, Application of optimal band-limited control protocols to quantum noise sensing, Nat. Commun. **8**, 2189 (2017).
[41] L. M. Norris, G. A. Paz-Silva, and L. Viola, Qubit noise spectroscopy for non-Gaussian dephasing environments, Phys. Rev. Lett. **116**, 150503 (2016).
[42] Z. Zhou, R. Sitler, Y. Oda, K. Schultz, and G. Quiroz, Quantum crosstalk robust quantum control, Phys. Rev. Lett. **131**, 210802 (2023).
[43] W. W. Ho and T. H. Hsieh, Efficient variational simulation of non-trivial quantum states, SciPost Phys. **6**, 029 (2019).
[44] M. S. Anis et al., Qiskit Textbook (Github, 2023), https://github.com/Qiskit/textbook.
[45] D. Zhu, N. M. Linke, M. Benedetti, K. A. Landsman, N. H. Nguyen, C. H. Alderete, A. Perdomo-Ortiz, N. Korda, A. Garfoot, C. Brecque, L. Egan, O. Perdomo, and C. Monroe, Training of quantum circuits on a hybrid quantum computer, Sci. Adv. **5**, aaw9918 (2019).
[46] M. Streif and M. Leib, Training the quantum approximate optimization algorithm without access to a quantum processing unit, Quantum Sci. Technol. **5**, 034008 (2020).
[47] M. M. Wauters, E. Panizon, G. B. Mbeng, and G. E. Santoro, Reinforcement-learning-assisted quantum optimization, Phys. Rev. Res. **2**, 033446 (2020).
[48] H. Ball and M. J. Biercuk, Walsh-synthesized noise filters for quantum logic, EPJ Quantum Technol. **2**, 11 (2015).
[49] R. Kubo, Generalized cumulant expansion method, J. Phys. Soc. Jpn. **17**, 1100 (1962).
[50] D. A. Lidar, P. Zanardi, and K. Khodjasteh, Distance bounds on quantum dynamics, Phys. Rev. A **78**, 012308 (2008).
[51] K. Khodjasteh and D. A. Lidar, Rigorous bounds on the performance of a hybrid dynamical-decoupling quantum-computing scheme, Phys. Rev. A **78**, 012355 (2008).
[52] G. S. Uhrig and D. A. Lidar, Rigorous bounds for optimal dynamical decoupling, Phys. Rev. A **82**, 012301 (2010).
[53] H. K. Ng, D. A. Lidar, and J. Preskill, Combining dynamical decoupling with fault-tolerant quantum computation, Phys. Rev. A **84**, 012305 (2011).
[54] J. C. Spall et al., Multivariate stochastic approximation using a simultaneous perturbation gradient approximation, IEEE Trans. Autom. Control **37**, 332 (1992).
[55] F. K. Wilhelm, S. Kirchhoff, S. Machnes, N. Wittler, and D. Sugny, An introduction into optimal control for quantum technologies, arXiv:2003.10132.
[56] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, Theory of trotter error with commutator scaling, Phys. Rev. X **11**, 011020 (2021).
[57] E. Farhi, J. Goldstone, S. Gutmann, and M. Sipser, Quantum Computation by Adiabatic Evolution, arXiv:quant-ph/0001106.
[58] C. Developers, quantumlib/Cirq, cirq version 0.9.1, https://zenodo.org/doi/10.5281/zenodo.4062499.
[59] F. Barahona, On the computational complexity of Ising spin glass models, J. Phys. A: Math. Gen. **15**, 3241 (1982).
[60] A. Lucas, Ising formulations of many NP problems, Front. Phys. **2**, 5 (2014).
[61] P. C. Lotshaw and T. S. Humble, QAOA dataset, https://code.ornl.gov/qci/qaoa-dataset-version1.
[62] P. C. Lotshaw, T. S. Humble, R. Herrman, J. Ostrowski, and G. Siopsis, Empirical performance bounds for quantum approximate optimization, Quantum Inf. Process. **20**, 403 (2021).
[63] J. R. McClean, J. Romero, R. Babbush, and A. Aspuru-Guzik, The theory of variational hybrid quantum-classical algorithms, New J. Phys. **18**, 023023 (2016).
[64] P. J. J. O'Malley, R. Babbush, I. D. Kivlichan, J. Romero, J. R. McClean, R. Barends, J. Kelly, P. Roushan, A. Tranter, N. Ding,

B. Campbell, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, A. G. Fowler, E. Jeffrey, E. Lucero, A. Megrant, J. Y. Mutus *et al.*, Scalable quantum simulation of molecular energies, Phys. Rev. X **6**, 031007 (2016).

[65] C. Dalyac, L. Henriet, E. Jeandel, W. Lechner, S. Perdrix, M. Porcheron, and M. Veshchezerova, Qualifying quantum approaches for hard industrial optimization problems. A case study in the field of smart-charging of electric vehicles, EPJ Quantum Technol. **8**, 12 (2021).

[66] S. Bravyi, A. Kliesch, R. Koenig, and E. Tang, Obstacles to variational quantum optimization from symmetry protection, Phys. Rev. Lett. **125**, 260505 (2020).

[67] E. Farhi, D. Gamarnik, and S. Gutmann, The quantum approximate optimization algorithm needs to see the whole graph: A typical case, arXiv:2004.09002.

[68] E. Farhi, D. Gamarnik, and S. Gutmann, The quantum approximate optimization algorithm needs to see the whole graph: Worst case examples, arXiv:2005.08747.

[69] M. L. Wall, M. R. Abernathy, and G. Quiroz, Generative machine learning with tensor networks: Benchmarks on near-term quantum computers, Phys. Rev. Res. **3**, 023010 (2021).

[70] M. L. Wall, P. Titum, G. Quiroz, M. Foss-Feig, and K. R. A. Hazzard, Tensor-network discriminator architecture for classification of quantum data on quantum computers, Phys. Rev. A **105**, 062439 (2022).

[71] M. L. Wall, A. Reilly, J. S. Van Dyke, C. Broholm, and P. Titum, Quantum tensor networks algorithms for evaluation of spectral functions on quantum computers, Phys. Rev. B **110**, 241402 (2024).

[72] P. Gokhale, Y. Ding, T. Propson, C. Winkler, N. Leung, Y. Shi, D. I. Schuster, H. Hoffmann, and F. T. Chong, Partial compilation of variational algorithms for noisy intermediate-scale quantum machines, in *Proceedings of the 52nd Annual IEEE/ACM International Symposium on Microarchitecture* (IEEE, New York, 2019), pp. 266–278.

[73] R. Barends, A. Shabani, L. Lamata, J. Kelly, A. Mezzacapo, U. L. Heras, R. Babbush, A. G. Fowler, B. Campbell, Y. Chen *et al.*, Digitized adiabatic quantum computing with a superconducting circuit, Nature (London) **534**, 222 (2016).

[74] Y. Nam, J.-S. Chen, N. C. Pisenti, K. Wright, C. Delaney, D. Maslov, K. R. Brown, S. Allen, J. M. Amini, J. Apisdorf *et al.*, Ground-state energy estimation of the water molecule on a trapped-ion quantum computer, npj Quantum Inf. **6**, 33 (2020).

[75] D. Rabinovich, E. Campos, S. Adhikary, E. Pankovets, D. Vinichenko, and J. Biamonte, Robustness of variational quantum algorithms against stochastic parameter perturbation, Phys. Rev. A **109**, 042426 (2024).

[76] DOE Public Access Plan, https://www.energy.gov/doe-public-access-plan

[77] R. F. Snider, Perturbation variation methods for a quantum Boltzmann equation, J. Math. Phys. **5**, 1580 (1964).